

Postprocessing of mixed elastic eigenvalues

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Abstract

This thesis is about improving the eigenvalue approximation of the Falk mixed eigenvalue method in elasticity with reduced symmetry using a local postprocessing. Assuming smooth eigenfunctions, the postprocessing yields an improvement in the approximation of eigenfunctions and eigenvalues. The well-posedness of the associated source problem and the stability of the corresponding Falk mixed finite element method is discussed. The absence of spurious solutions is shown, and a priori error estimates for the eigenvalue and eigenfunction approximation are presented. Using the postprocessed eigenfunction, a reliable and efficient a posteriori error estimator for the postprocessed eigenvalue approximation error is proposed. For smooth eigenvalues and eigenfunctions, numerical experiments support the higher-order convergence of the postprocessed eigenvalues. For the postprocessed lowest-order Falk finite element eigenvalue approximation, we even observe superconvergence on some meshes. On non-convex domains, where smooth eigenvalues and eigenfunctions cannot be expected, adaptive mesh refinement yields optimal higher convergence in the eigenvalue approximation.

1 Introduction

In the context of elasticity, eigenvalue problems are, for example, in structural mechanics of great interest. Eigenfrequencies are certain frequencies that cause an elastic body to tend to vibrate. Some eigenfrequencies cause an elastic body to deform into a specific shape. These shapes are called eigenmodes. Therefore, in structural engineering, it is essential to analyse eigenfrequencies, which motivates to study corresponding finite element methods.

The classical displacement approach in elasticity has some disadvantages compared to mixed finite element methods. By mixed finite element methods, we mean methods in which both the displacement vector and the stress tensor are calculated simultaneously using the Hellinger-Reissner principle. This allows a direct calculation of stresses, which is often desirable and not in a second step using the gradient of displacements. In addition, displacement methods lead to the Locking effect, which can cause a degeneration of the solution in the case of incompressible or nearly incompressible materials. On the other hand, for mixed methods, it is known that a number of suitable finite elements lead to good approximations also for incompressible or nearly incompressible materials.

In the context of mixed problems (saddle point problems), Brezzi [9] provides a framework for well-posedness. It turns out that the main part is the inf-sup condition, which is also called Ladyženskaya-Babuška-Brezzi condition (LBB-condition).

The stability for mixed methods in elasticity is more difficult than for displacement methods. Appropriate mixed methods usually satisfy the equilibrium condition, which is a basic property in solid mechanics. Here, the normal component of the stress tensor is assumed to be continuous at the edges of the elements.

Due to conservation of angular momentum, the stress tensor is symmetric. Methods that strongly demand this property are highly non-trivial and require many degrees of freedom to ensure continuity of the stress along interelement boundaries. This motivates the introduction of a Lagrange multiplier and ensure symmetry only in average, which was first presented by Fraijs de Veubeke [26]. This method provides solutions for the stress tensor σ , the displacement u , and the skew-symmetric part

γ simultaneously. An approach for the unified analysis of stable finite elements in elasticity with weakly imposed symmetry is proposed by Lee [21].

The basic reference for compact eigenvalue problems is the Babuška-Osborn theory. In this context, no spurious eigenvalues pollute the spectrum and the associated discrete eigenspaces converge to the continuous ones with correct multiplicity. However, for mixed problems, the solution operators are generally not compact and the analysis requires a different theory. Mercier et al. [22] developed a corresponding framework for mixed problems. The conditions, cf. Boffi [3], are closely related to the stability of the problem. For simplicity, we will restrict ourselves to simple eigenvalues. Here, the associated eigenspaces are one-dimensional.

This thesis aims to improve the Falk finite element eigenvalue approximation using a local postprocessing. To this end, it is important that we can decouple the approximation error of the eigenfunction, and the approximation error of the stress and the skew-symmetric part. Approximating the stress tensor and the skew-symmetric part with polynomials of degree k , and the eigenfunctions with polynomials of degree $k - 1$ yields

$$\begin{aligned}\|\sigma - \sigma_h\|_0 + \|\gamma - \gamma_h\|_0 &\leq Ch^{k+1}(|\sigma|_{k+1} + |\gamma|_{k+1}), \\ \|u - u_h\|_0 &\leq Ch^k(|u|_k + |\sigma|_k + |\gamma|_k),\end{aligned}\tag{1}$$

provided the solutions σ , u , γ are sufficiently smooth. These are estimates valid for the source problem. Using a duality argument (Aubin-Nitsche trick), we have shown the following convergence result for the source problem:

$$\|P_h u - u_h\|_0 = \mathcal{O}(h^{k+2}).\tag{2}$$

This is valid except for the lowest-order Falk finite element approximation, where we have one order of convergence less. Moreover, the eigenvalue problem requires a slightly different analysis. For the Stokes eigenvalue problem, Gedicke and Khan [17] have proven an estimate of the form (2). With the necessary modifications, we will show this estimate for the eigenvalue problem in elasticity with reduced symmetry.

The results (1) and (2) motivate a local postprocessing in the spirit of Stenberg [24] to improve the eigenfunction approximation. The particular postprocessing that we apply was first suggested by Gopalakrishnan and Guzmán [18]. Then, the eigenvalue approximation

$$|\lambda - \lambda_h| = \mathcal{O}(h^{2k})$$

can be improved using the postprocessed eigenfunction u_h^* . The resulting postprocessed eigenvalue approximation yields

$$|\lambda - \lambda_h^*| = \mathcal{O}(h^{2k+2}).$$

We propose a reliable and efficient a posteriori error estimator η for the postprocessed eigenfunction approximation in the spirit of Carstensen [11]:

$$\begin{aligned}\eta^2 &= \|\mathbb{C}^{-1}\sigma_h + \gamma_h - \nabla_{\mathcal{T}_h} u_h^*\|_0^2 + \sum_{T \in \mathcal{T}_h} h_T^2 \|\lambda_h^* u_h^* + \operatorname{div} \sigma_h\|_{0,T}^2 \\ &\quad + \sum_{E \in \mathcal{E}_h} h_E^{-1} \|[u_h^*]\|_{0,E}^2 + \|\operatorname{skw}(\sigma_h)\|_0^2.\end{aligned}$$

For the Stokes eigenvalue problem, similarly, an error estimator has been presented by Gedicke and Khan [17]. In the construction, we use the primal mixed method

in elasticity with weakly imposed symmetry. The well-posedness of this problem does not follow directly by the well-posedness of the primal mixed problem in elasticity with symmetric stress. Here, we show the inf-sup condition using Helmholtz-decomposition.

We use an error estimator μ for the eigenvalue approximation $|\lambda - \lambda_h|$, which we obtain without proof from an estimator presented in Carstensen and Dolzmann [12].

For the lowest-order Falk element, the analysis shows that we can only expect one order of convergence less. Indeed, we observe a better convergence rate in all of our experiments. Moreover, we observe exceptional superconvergence on some meshes due to cancellation.

The thesis is organized as follows: in section 2, we will introduce some necessary notations first. Then the model problem in linear elasticity and the mixed formulation with weakly imposed symmetry are stated. Moreover, the well-posedness in this context is discussed.

Section 3 is about stable finite elements in the weakly symmetric elasticity context. These elements are indeed closely related to Stokes stable elements. Further, necessary polynomial spaces are introduced and Falk's element [1], which we use throughout, is discussed.

In section 4, the eigenvalue problem is stated and the theory for mixed eigenvalue problems in the sense of Boffi [3, Chapter 14] is adapted to our problem. An identity for the eigenvalue approximation error is established and a priori error estimates for the eigenvalue approximation and the eigenfunction approximation are presented.

Section 5 continues with the local postprocessing and its higher-order convergence analysis. Further, an identity for the postprocessed eigenvalue is derived, and the improved eigenvalue approximation is shown.

For the postprocessed eigenvalue, a reliable and efficient a posteriori error estimator is proposed in section 6. The higher-order convergence of the postprocessed Falk finite element eigenvalue approximation is presented in the numerical experiments on three different domains in section 7.

2 Notations and Preliminaries

First, let us introduce some notations. The space of square integrable (classes of measurable) functions on $\Omega \subset \mathbb{R}^2$ is given by

$$L^2(\Omega) = \{v \in \Omega : \|v\|_{L^2(\Omega)}^2 := \int_{\Omega} |v(x)|^2 dx < \infty\}.$$

To simplify the notation, we write

$$\|\cdot\|_0 := \|\cdot\|_{L^2(\Omega)}.$$

This space is only defined up to null sets, which are sets of measure zero. This means for $u, v \in L^2(\Omega)$, it holds

$$\|u\|_0 = \|v\|_0 \Rightarrow u = v \text{ almost everywhere.}$$

We will neglect to write almost everywhere in the following and, for simplicity, let us just say functions to members of this space. The L^2 -space is a Hilbert space with inner product

$$(u, v) := \int_{\Omega} uv \, dx$$

for $u, v \in L^2(\Omega)$. Further, for $s \geq 0$, we introduce Sobolev spaces by

$$H^s(\Omega) = \{v \in L^2(\Omega) : \|v\|_{H^s(\Omega)}^2 < \infty\}$$

with the associated norm

$$\|v\|_s := \|v\|_{H^s(\Omega)} = \sum_{j=0}^s \|D^j v\|_0$$

and corresponding seminorm

$$|v|_s := |v|_{H^s(\Omega)} = \|D^s v\|_0$$

for $v \in H^s(\Omega)$. Further, let us introduce

$$H^{\text{div}}(\Omega) = \{v \in L^2(\Omega) : \text{div } v \in L^2(\Omega)\}$$

with corresponding norms

$$\begin{aligned} \|u\|_{\text{div}} &:= \|\text{div } u\|_0, \\ \|u\|_{H^{\text{div}}} &:= \|u\|_0 + \|u\|_{\text{div}}, \end{aligned}$$

for $u \in H^{\text{div}}(\Omega)$.

We extend the definition of $L^2(\Omega)$ to vector-valued and matrix-valued function spaces and denote them by $L^2(\Omega; \mathbb{R}^2)$ and $L^2(\Omega; \mathbb{R}^{2 \times 2})$. This can analogously be done for the remaining introduced spaces. Let

$$v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \in \mathbb{R}^2, \quad \tau = \begin{pmatrix} \tau_{11} & \tau_{12} \\ \tau_{21} & \tau_{22} \end{pmatrix} \in \mathbb{R}^{2 \times 2}, \quad \sigma = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix} \in \mathbb{R}^{2 \times 2},$$

and suppose v and τ are sufficiently smooth. Then, we define the gradient by

$$\nabla v := \begin{pmatrix} \partial_1 v_1 & \partial_2 v_1 \\ \partial_1 v_2 & \partial_2 v_2 \end{pmatrix},$$

the divergence by

$$\operatorname{div} v := \partial_1 v_1 + \partial_2 v_2, \quad \operatorname{div} \tau := \begin{pmatrix} \partial_1 \tau_{11} + \partial_2 \tau_{12} \\ \partial_1 \tau_{21} + \partial_2 \tau_{22} \end{pmatrix},$$

and the two curl operators by

$$\operatorname{curl} v := -\partial_2 v_1 + \partial_1 v_2, \quad \operatorname{Curl} v := \begin{pmatrix} -\partial_2 v_1 & \partial_1 v_1 \\ -\partial_2 v_2 & \partial_1 v_2 \end{pmatrix}.$$

The matrix product and the trace operator are defined by

$$\tau : \sigma := \sum_{i,j=1}^2 \tau_{ij} \sigma_{ij}, \quad \operatorname{tr}(\tau) := \sum_{j=1}^2 \tau_{jj}.$$

The space of square integrable skew-symmetric tensors is given by

$$\mathbf{L}_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}) := \{\tau \in \mathbf{L}^2(\Omega, \mathbb{R}^{2 \times 2}) : \tau^\top = -\tau\},$$

the skew-symmetric part of $\tau \in \mathbb{R}^{2 \times 2}$ by

$$\operatorname{skw} \tau := \frac{1}{2}(\tau - \tau^\top) = \frac{1}{2} \begin{pmatrix} 0 & \tau_{12} - \tau_{21} \\ \tau_{21} - \tau_{12} & 0 \end{pmatrix},$$

and the symmetric part by

$$\operatorname{sym} \tau := \tau - \operatorname{skw} \tau := \frac{1}{2}(\tau + \tau^\top).$$

Moreover, define

$$\operatorname{as} \tau = \frac{1}{2}(\tau_{12} - \tau_{21}).$$

For any $q \in \mathbf{L}^2(\Omega)$, let

$$\mathbf{S}^2(q) := \begin{pmatrix} 0 & q \\ -q & 0 \end{pmatrix} \in \mathbf{L}_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}).$$

Throughout, let $\Omega \in \mathbb{R}^2$ be a convex, polygonal, and bounded Lipschitz domain. Suppose the boundary of Ω can be split up into a Dirichlet part Γ_D and a Neumann part Γ_N such that

$$\partial\Omega = \Gamma_D \cup \Gamma_N.$$

Triangulation

Let $\mathcal{T}_h$ be a regular triangulation of Ω into triangles T with positive area, i.e.

$$\bar{\Omega} = \bigcup_{T \in \mathcal{T}_h} T,$$

and any two elements $T_i, T_j \in \mathcal{T}_h$ that are distinct $T_i \neq T_j$ have either exactly one common edge, share exactly one common node, or the intersection $T_i \cap T_j = \emptyset$ is empty. The set of all edges in the triangulation is denoted by

$$\mathcal{E} := \mathcal{E}_I \cup \mathcal{E}_D \cup \mathcal{E}_N,$$

where $\mathcal{E}_I$ is the set of interior element edges, $\mathcal{E}_D$ is the set of edges on the Dirichlet boundary Γ_D , and $\mathcal{E}_N$ is the set of edges on the Neumann boundary Γ_N . By assuming that the domain is polygonal, the analysis in the further discussion is much simpler and less technical. There are no curved edges, and the boundary only contains edges of triangles.

In the following, let $\mathcal{T}_h$ be a uniform triangulation, i.e. there exists a positive number κ , and each element $T \in \mathcal{T}_h$ contains a circle with radius ρ_T such that

$$\rho_T \geq \frac{h}{\kappa}$$

with $h = \max_{T \in \mathcal{T}_h} h_T$. Even though the statements in the following can be extended to quasiuniform triangulations without great effort, we will restrict ourselves to uniform triangulations for the sake of simplicity.

Traces

It is known that evaluations of L^2 -functions are not explained on the boundary $\partial\Omega$. In fact, one can change the value of the function on $\partial\Omega$ arbitrarily without changing the function in terms of the L^2 -space. This is because $\partial\Omega$, as a subset of $\mathbb{R}^2$, is a set of measure zero.

On Sobolev spaces, evaluations of functions on the boundary are defined via the trace operator

$$\begin{aligned} \gamma_{\text{tr}} : H^1(\Omega; \mathbb{R}^2) &\rightarrow L^2(\partial\Omega), \\ v &\mapsto \gamma_{\text{tr}}(v), \end{aligned}$$

which is a linear, continuous and surjective operator. We write $v|_{\partial\Omega}$ for evaluations on the boundary of $v \in H^1(\Omega)$ in the following, i.e.

$$v|_{\partial\Omega} = \gamma_{\text{tr}} v.$$

Furthermore, for any measurable subset $\Gamma \in \partial\Omega$, we can evaluate $v|_{\Gamma}$.

It is still not clear how to interpret normal components of H^{div} -functions on the boundary. Here we need different spaces $H^{1/2}$ and $H^{-1/2}(\partial\Omega)$. For any $v \in H^{\text{div}}(\Omega; \mathbb{R}^2)$, we can define $q \cdot n|_{\partial\Omega} \in H^{-1/2}(\partial\Omega)$. Green's identity states

$$\int_{\Omega} q \operatorname{div} v \, dx = - \int_{\Omega} \nabla q \cdot v \, dx + \int_{\partial\Omega} q v \cdot n \, ds$$

for any $q \in H^1(\Omega)$. The trace operator is defined by

$$\begin{aligned} \gamma_{\text{tr}} : H^{\text{div}}(\Omega; \mathbb{R}^2) &\rightarrow H^{-1/2}(\partial\Omega), \\ v &\mapsto \gamma_{\text{tr}}(v), \end{aligned}$$

so that the normal component of an H^{div} -function can be explained as a continuous linear functional on $H^{1/2}(\partial\Omega)$.

Material laws and connection to Stokes

In the elasticity context, the function $u : \Omega \rightarrow \mathbb{R}^2$ represents the displacement and $\sigma : \Omega \rightarrow \mathbb{R}^{2 \times 2}$ the symmetric stress tensor. The linear Green strain

$$\epsilon(u) = \frac{1}{2}(\nabla u + \nabla u^\top)$$

is defined as the symmetrised gradient of u and the isotropic fourth-order material tensor is given by

$$\mathbb{C}\epsilon(u) := \lambda \operatorname{tr}(\epsilon(u)) \mathbf{I}_{2 \times 2} + 2\mu \epsilon(u),$$

where λ and μ are the two positive Lamé parameters. Let us note $\mathbb{C}$ maps $\mathbb{R}^{2 \times 2}$ into itself.

Due to Hooke's law, the stress σ and the strain ϵ are proportional in Ω depending on the material, which is prescribed by λ and μ , i.e.

$$\sigma = \mathbb{C}\epsilon(u). \quad (3)$$

The rigid body motions are given by the elements of

$$\text{RM} = \left\{ v : v = \begin{pmatrix} a \\ b \end{pmatrix} + c \begin{pmatrix} x_2 \\ -x_1 \end{pmatrix} \text{ with } a, b, c \in \mathbb{R} \right\},$$

and it is easy to verify that

$$\begin{aligned} v \in \text{RM} &\Leftrightarrow \epsilon(v) = 0, \\ v \in \text{RM} &\Rightarrow \operatorname{div} v = 0. \end{aligned}$$

Obviously, by (3), rigid body motions do not influence the stress. By taking the trace on both sides in (3), we have

$$\operatorname{tr}(\sigma) = \operatorname{tr}(\mathbb{C}\epsilon(u)) = (n\lambda + 2\mu) \operatorname{tr}(\epsilon(u)),$$

where n is the dimension of the space, which in our case is $n = 2$. The operator $\mathbb{C}$ is invertible with inverse

$$\epsilon(u) = \mathbb{C}^{-1}\sigma = \frac{1}{2\mu} \left(\sigma - \frac{\lambda}{2(\mu + \lambda)} \operatorname{tr} \sigma \mathbf{I}_{2 \times 2} \right). \quad (4)$$

Incompressible materials do not change their volume under pressure. They only change their shape. In fluid dynamics, this would mean that in any closed region $K \subset \Omega$ the inflow must balance the outflow across the boundary. Therefore applying the divergence theorem, we have

$$0 = \int_{\partial K} u \cdot n \, dx = \int_K \operatorname{div} u \, dx.$$

Because this holds for any closed $K \subset \Omega$, this implies

$$\operatorname{div} u = 0 \quad \text{in } \Omega.$$

By taking the trace in (4) on both sides, we have

$$0 = \operatorname{div} u = \operatorname{tr}(\epsilon(u)) = \frac{1}{2\mu} \left(\operatorname{tr} \sigma - \frac{\lambda}{\mu + \lambda} \operatorname{tr} \sigma \right) = \frac{1}{2(\lambda + \mu)} \operatorname{tr} \sigma$$

and the incompressible case corresponds to $\lambda \rightarrow \infty$. Further, (4) is still meaningful and the corresponding limit is

$$\frac{1}{2\mu}(\sigma - \frac{1}{2} \text{tr } \sigma \text{I}_{2 \times 2}) = \frac{1}{2\mu} \text{dev } \sigma.$$

Here we see the connection between the elasticity equations and Stokes equations, i.e. for Stokes, we have

$$\epsilon(u) = \frac{1}{2\mu} \text{dev } \sigma.$$

For any 2×2 tensor τ , its deviator $\text{dev } \tau$ is defined by

$$\text{dev } \tau := \tau - \frac{1}{2} \text{tr}(\tau) \text{I}_{2 \times 2}$$

and is trace free. Let us define the energy norm

$$\|\tau\|_{\mathbb{C}^{-\frac{1}{2}}} := \|\mathbb{C}^{-\frac{1}{2}}\tau\|_0 = (\mathbb{C}^{-\frac{1}{2}}\tau, \mathbb{C}^{-\frac{1}{2}}\tau) = (\mathbb{C}^{-1}\tau, \tau) = (\tau, \mathbb{C}^{-1}\tau), \quad (5)$$

which is a weighted L^2 -norm. The last equation above is valid due to

$$(\mathbb{C}^{-1}\sigma, \tau) = \int_{\Omega} \frac{1}{2\mu} \sigma : \tau - \frac{\lambda}{4\mu(\lambda + \mu)} (\sigma_{11} + \sigma_{22})(\tau_{11} + \tau_{22}) \, dx = (\sigma, \mathbb{C}^{-1}\tau). \quad (6)$$

Kinematik and strong formulation

In the elasticity context, $\Omega \subset \mathbb{R}^2$ is an elastic body. The body Ω is loaded with applied volume forces $f \in L^2(\Omega; \mathbb{R}^2)$ and some surface traction $g \in L^2(\Gamma_N; \mathbb{R}^2)$ acting on the Neumann part Γ_N of the boundary $\partial\Omega$ with $|\Gamma_N| > 0$.

For the moment, let us assume u and σ are smooth enough to state the strong formulation in linear elasticity:

$$\begin{aligned} \mathbb{C}^{-1}\sigma &= \epsilon(u) && \text{in } \Omega, \\ \text{div } \sigma + f &= 0 && \text{in } \Omega, \\ \sigma &= \sigma^\top && \text{in } \Omega, \\ u &= 0 && \text{on } \Gamma_D, \\ \sigma \cdot n &= g && \text{on } \Gamma_N, \end{aligned} \quad (7)$$

with n being the unit normal pointing in outward direction. The first three equations are derived from physical axioms. Due to conservation of energy, we have Hooke's law, which is the first equation. The second equation is obtained from conservation of momentum, which is described by Newton's second law. For any particle x_i moving across Ω over time, the change of momentum of x_i equals $m_i \cdot x_i''(t)$. By Newton's second law, we have

$$m_i \cdot x_i''(t) = f_i(t) + \sum_{j \neq i} f_{ij}(t),$$

where $f_i(t)$ is the external force acting on $x_i(t)$ at time t and $f_{ij}(t)$ the force of another particle x_j that act on x_i at time t .

Let $U \subset \Omega$ be a deformation. The rate of change of total momentum in U equals the forces that act on U . Since we consider a steady state, the deformation is constant over time and the rate of change of total momentum in U is zero, i.e.

$$0 = \int_{\partial U} \sigma \cdot n \, dx - \int_U f \, dx.$$

The first term describes surface forces, e.g. due to traction or pressure differences, and the second term describes volume forces, e.g. due to a load acting on the body. Applying the divergence theorem, we have

$$\int_U f + \operatorname{div} \sigma \, dx = 0$$

for any $U \subset \Omega$ and therefore $f + \operatorname{div} \sigma = 0$ in Ω . Due to conservation of mass and conservation of angular momentum, the stress σ is symmetric

$$\sigma = \sigma^T.$$

It is not easy to construct finite elements in which this condition is built-in. The Arnold-Winther family provides such an element but will not be discussed further here. To avoid this difficulty, a weaker symmetry condition can be fulfilled using a Lagrange multiplier.

Weak formulation in linear elasticity

First, let us define the spaces

$$\begin{aligned} \Sigma_g &:= \{\tau \in L^2(\Omega; \mathbb{R}^{2 \times 2}) : \operatorname{div} \tau \in L^2(\Omega; \mathbb{R}^2), \tau \cdot n = g \text{ on } \Gamma_N\}, \\ U &:= L^2(\Omega; \mathbb{R}^2), \end{aligned}$$

and enforce symmetry of the stress strongly. The displacement-stress formulation in linear elasticity reads as follows: We seek for functions $(u, \sigma) \in U \times \Sigma_g \cap \{\sigma : \sigma = \sigma^T\}$ such that

$$\begin{aligned} \int_{\Omega} \mathbb{C}^{-1} \sigma : \tau \, dx + \int_{\Omega} \operatorname{div} \tau \cdot u \, dx &= 0 & \text{for all } \tau \in \Sigma_0 \cap \{\sigma : \sigma = \sigma^T\}, \\ \int_{\Omega} \operatorname{div} \sigma \cdot v \, dx &= - \int_{\Omega} f \cdot v \, dx & \text{for all } v \in U. \end{aligned}$$

While the Neumann boundary condition

$$\sigma \cdot n = g \quad \text{on } \Gamma_N$$

must strongly be included into the Sobolev space, the Dirichlet boundary condition

$$u = 0 \quad \text{on } \Gamma_D$$

appears naturally. In the first equation, we test with test functions τ that have homogeneous Neumann boundary conditions so that the boundary term that appears due to integration by parts vanishes. The equation

$$\operatorname{div} \sigma = f$$

is enforced by a Lagrange multiplier.

Weak formulation in linear elasticity with weakly imposed symmetry and well-posedness

From now on, we no longer assume σ to be symmetric. Using a Lagrange multiplier, we enforce a weaker assumption, namely symmetry only in average. The gradient of u can be split up into its symmetric and its skew-symmetric part

$$\nabla u = \epsilon(u) + \text{skw}(\nabla u).$$

By Hooke's law, we have

$$\nabla u = \mathbb{C}^{-1}\sigma + \text{skw}(\nabla u).$$

Let us introduce a new unknown γ , which represents the skew-symmetric of the gradient of u . Then the new strong formulation in linear elasticity reads

$$\begin{aligned} \mathbb{C}^{-1}\sigma - \nabla u + \gamma &= 0 & \text{in } \Omega, \\ \text{div } \sigma + f &= 0 & \text{in } \Omega, \\ \sigma - \sigma^\top &= 0 & \text{in } \Omega, \\ u &= 0 & \text{on } \Gamma_D, \\ \sigma \cdot n &= g & \text{on } \Gamma_N. \end{aligned} \tag{8}$$

The space of square integrable skew-symmetric tensors is given by

$$X := \{\eta \in L^2(\Omega; \mathbb{R}^{2 \times 2}) : \text{skw}(\eta) = \eta\} \tag{9}$$

and let us define the forms

$$a : \Sigma_g \times \Sigma_g \rightarrow \mathbb{R}, \quad b : \Sigma_g \times U \rightarrow \mathbb{R}, \quad c : X \times \Sigma_g \rightarrow \mathbb{R},$$

by

$$a(\sigma, \tau) := \int_{\Omega} \mathbb{C}^{-1}\sigma : \tau \, dx, \quad b(\tau, v) := \int_{\Omega} \text{div } \tau \cdot v \, dx, \quad c(\eta, \tau) := \int_{\Omega} \eta : \tau \, dx. \tag{10}$$

For the moment, we assume u is smooth and $u = 0$ on Γ_D . Further, let $\tau \in \Sigma_0$. Then with $\nabla u = \mathbb{C}^{-1}\sigma + \gamma$, we have

$$\begin{aligned} 0 &= \int_{\Gamma_D} u \cdot \tau \cdot n \, dx + \int_{\Gamma_N} u \cdot \tau \cdot n \, dx = \int_{\partial\Omega} u \cdot \tau \cdot n \, dx \\ &= \int_{\Omega} \nabla u : \tau \, dx + \int_{\Omega} u \cdot \text{div } \tau \, dx \\ &= \int_{\Omega} \mathbb{C}^{-1}\sigma : \tau \, dx + \int_{\Omega} u \cdot \text{div } \tau \, dx + \int_{\Omega} \gamma : \tau \, dx. \end{aligned}$$

The dual weak formulation in linear elasticity with weakly imposed symmetry reads as follows: find $(\sigma, u, \gamma) \in \Sigma_g \times U \times X$ such that

$$\begin{aligned} a(\sigma, \tau) + b(\tau, u) + c(\gamma, \tau) &= 0 & \text{for all } \tau \in \Sigma_0, \\ b(\sigma, v) &= -(f, v) & \text{for all } v \in U, \\ c(\sigma, \eta) &= 0 & \text{for all } \eta \in X. \end{aligned} \tag{11}$$

Due to linearity of the inverse material tensor $\mathbb{C}^{-1}$ and the divergence operator, all three forms $a(\bullet, \bullet)$, $b(\bullet, \bullet)$, and $c(\bullet, \bullet)$ are bilinear. Moreover, the identities

$$\sigma : \text{tr } \tau \, \mathbb{I}_{2 \times 2} = \text{tr } \sigma \, \text{tr } \tau, \quad \text{and} \quad \text{dev } \sigma : \text{dev } \tau = \sigma : \tau - \frac{1}{2} \text{tr } \sigma \, \text{tr } \tau, \quad (12)$$

whose validity can be easily proven, may be used to show that $a(\bullet, \bullet)$ can be reformulated to

$$a(\sigma, \tau) = \int_{\Omega} \left(\frac{1}{2\mu} \text{dev } \sigma : \text{dev } \tau + \frac{1}{4(\lambda + \mu)} \text{tr}(\sigma) \text{tr}(\tau) \right) dx.$$

By (5), we observe that $a(\bullet, \bullet)$ is symmetric, and furthermore

$$a(\tau, \tau) \geq 0 \quad \text{for all } \tau \in \Sigma_g.$$

By Brezzi [9], the standard conditions for well-posedness of saddle point problems are the inf-sup condition or sometimes called Ladyženskaya-Babuška-Brezzi (LBB) condition, and a coercivity condition. Well-posedness in the sense that there exists a unique solution $(\sigma, u, \gamma) \in \Sigma_g \times U \times X$ satisfying the equations (11) and the solution depends continuously on the given data. Let $L : \Sigma_g \times U \times X \rightarrow \Sigma'_g \times U' \times X'$ be given by

$$L(\tau, v, \eta) = (0, -f, 0).$$

Brezzi's Splitting Theorem states that L is an isomorphism if the following conditions are satisfied: there exist constants $C_1, C_2 \geq 0$ such that

$$\inf_{\substack{v \in U \\ \eta \in X}} \sup_{\tau \in \Sigma_g} \frac{b(\tau, v) + c(\tau, \eta)}{\|v\|_0 + \|\eta\|_0} \geq C_1 \|\tau\|_{\text{H}^{\text{div}}(\Omega)}, \quad (13)$$

and

$$a(\tau, \tau) \geq C_2 \|\tau\|_{\text{H}^{\text{div}}(\Omega)}^2 \quad \text{for all } \tau \in \Sigma_g \cap \ker(B + C). \quad (14)$$

The following proposition ensures the inf-sup condition (13).

Proposition 2.1. *[5, Proposition 2] For any $v \in U$ and $\eta \in X$, there exists a tensor $\tau \in \Sigma_g$ satisfying*

$$b(\tau, v) + c(\tau, \eta) = \|v\|_0^2 + \|\eta\|_0^2.$$

Moreover, there exists a constant $C \geq 0$ such that

$$\|\tau\|_{\text{H}^{\text{div}}(\Omega)} \leq C(\|v\|_0 + \|\eta\|_0).$$

By (12), it is easy to see that the inequalities

$$\|\text{dev } \tau\|_0 \leq \|\tau\|_0 \quad \text{and} \quad \|\tau\|_0 \geq C \|\text{tr } \tau\|_0$$

hold. In the discussion about the coercivity condition (14), it is important that this holds uniformly in λ , so that the well-posedness of the problem does not break down in the incompressible or nearly incompressible case. For any $\tau \in \text{H}^{\text{div}}(\Omega; \mathbb{R}^{2 \times 2})$, we have

$$\begin{aligned} a(\tau, \tau) &= \frac{1}{2\mu} \int_{\Omega} \tau : \tau - \frac{\lambda}{2(\lambda + \mu)} (\text{tr } \tau)^2 dx \geq \frac{1}{2\mu} \|\tau\|_0^2 - \frac{\lambda}{2\mu(\lambda + \mu)} \|\tau\|_0^2 \\ &\geq \frac{1}{2(\lambda + \mu)} \|\tau\|_0^2. \end{aligned}$$

Then the constant $C_2 = C_2(\lambda, \mu)$ depends on the Lamé parameter λ and whenever λ tends to infinity, we have

$$\lim_{\lambda \rightarrow \infty} a(\tau, \tau) \rightarrow 0.$$

Hence coercivity of $a(\bullet, \bullet)$ w.r.t. $\|\bullet\|_{\text{H}^{\text{div}}}$ does not hold. We can get around this issue, because we only need coercivity in the kernel of the second and the third form. Let us define the operators $B : \Sigma_g \rightarrow U'$ and $C : \Sigma_g \rightarrow X'$ by

$$\begin{aligned} (B\tau)(v) &:= (v, \text{div } \tau) & \text{for all } v \in U, \\ (C\tau)(\eta) &:= (\eta, \tau) & \text{for all } \eta \in X. \end{aligned} \tag{15}$$

The kernel of $B + C$ is given by

$$\ker(B + C) = \{\tau \in \Sigma_g : b(\tau, v) + c(\eta, \tau) = 0 \ \forall v \in U, \forall \eta \in X\},$$

and contains symmetric tensors τ that, in particular, satisfy

$$\text{div } \tau = 0.$$

Let us define the space of square integrable 2×2 tensors that are trace free in average by

$$\text{L}_0^2(\Omega; \mathbb{R}^{2 \times 2}) := \{\tau \in \text{L}^2(\Omega; \mathbb{R}^{2 \times 2}) : \int_{\Omega} \text{tr } \tau \, dx = 0\}.$$

This space is isometric to the space of non-constant on the diagonal components and square integrable 2×2 tensors $\text{L}^2(\Omega; \mathbb{R}^{2 \times 2})/(\mathbb{R} \text{I}_{2 \times 2})$. Indeed, there exists an isometric isomorphism between these two spaces.

Lemma 2.2. *[5, Proposition 1] There is a positive constant C such that for any $\tau \in \Sigma_g \cap \text{L}_0^2(\Omega; \mathbb{R}^{2 \times 2})$, the following estimate*

$$\|\tau\|_0 \leq C(\|\text{dev } \tau\|_0 + \|\text{div } \tau\|_0)$$

holds.

The statement of Lemma 2.2 ensures coercivity of the bilinear form $a(\bullet, \bullet)$ in a smaller space independent of λ . Let us remark that the deviator of $\mathbb{C}^{-1}\tau$ can be reformulated

$$\text{dev}(\mathbb{C}^{-1}\tau) = \mathbb{C}^{-1}\tau - \frac{1}{2} \text{tr}(\mathbb{C}^{-1}\tau) \text{I}_{2 \times 2} = \frac{1}{2\mu} \text{dev } \tau.$$

For any $\tau \in \ker(B + C) \cap \text{L}_0^2(\Omega; \mathbb{R}^{2 \times 2})$, we have

$$\begin{aligned} a(\tau, \tau) &= (\mathbb{C}^{-1}\tau, \tau) \geq (\text{dev}(\mathbb{C}^{-1}\tau), \text{dev } \tau) = \frac{1}{2\mu} \|\text{dev } \tau\|_0^2 \\ &\geq \frac{1}{2\mu} \|\tau\|_0^2 = \frac{1}{2\mu} \|\tau\|_{\text{H}^{\text{div}}}^2. \end{aligned} \tag{16}$$

The first inequality holds by (12) and the last by Lemma 2.2.

3 Finite elements in linear elasticity with weakly imposed symmetry

In this section, we will first discuss the discrete source problem. Stability of finite elements are closely related to Stokes stable elements here. For this purpose, an auxiliary Stokes problem is stated and its weak formulation briefly discussed. Finally, the Falk element is presented.

3.1 The construction of stable finite elements in this context

To solve the discrete variational problem to (11), we need appropriate finite dimensional subspaces $\Sigma_{g,h} \subset \Sigma_g$, $U_h \subset U$ and $X_h \subset X$. Then we seek for $(\sigma_h, u_h, \gamma_h) \in \Sigma_{g,h} \times U_h \times X_h$ satisfying

$$\begin{aligned} a(\sigma_h, \tau_h) + b(\tau_h, u_h) + c(\gamma_h, \tau_h) &= 0 & \text{for all } \tau_h \in \Sigma_{0,h}, \\ b(\sigma_h, v_h) &= -(f, v_h) & \text{for all } v_h \in U_h, \\ c(\gamma_h, \eta_h) &= 0 & \text{for all } \eta_h \in X_h. \end{aligned} \quad (17)$$

Let us define corresponding discrete operators $B_h : \Sigma_{g,h} \rightarrow U'_h$ and $C_h : \Sigma_{g,h} \rightarrow X'_h$ to (15) by

$$\begin{aligned} (B_h \tau_h)(v_h) &= b_h(\tau_h, v_h) & \text{for all } v_h \in U_h, \\ (C_h \tau_h)(\eta_h) &= c_h(\eta_h, \tau_h) & \text{for all } \eta_h \in X_h. \end{aligned}$$

The kernel of $B_h + C_h$ is given by

$$\ker(B_h + C_h) = \{\tau_h \in \Sigma_{g,h} : b(\tau_h, v_h) + c(\eta_h, \tau_h) = 0 \forall v_h \in U_h \text{ and } \forall \eta_h \in X_h\}. \quad (18)$$

To have a well-posed and stable discrete problem, we need suitable discrete conditions similar to (13) and (14). The "inclusion of the kernel" property

$$\ker(B_h + C_h) \subset \ker(B + C) \quad (19)$$

ensures coercivity of $a(\bullet, \bullet)$ w.r.t. $\|\bullet\|_{\text{H}^{\text{div}}}$. Indeed, for any $\tau_h \in \ker(B_h + C_h) \cap L_0^2(\Omega; \mathbb{R}^{2 \times 2}) \subset \ker(B + C) \cap L_0^2(\Omega; \mathbb{R}^{2 \times 2})$ the following holds:

$$\begin{aligned} a(\tau_h, \tau_h) &= (\mathbb{C}^{-1} \tau_h, \tau_h) \geq (\text{dev } \mathbb{C}^{-1} \tau_h, \text{dev } \tau_h) = \frac{1}{2\mu} \|\text{dev } \tau_h\|_0 \\ &\geq \frac{1}{2\mu} \|\tau_h\|_0 = \frac{1}{2\mu} \|\tau_h\|_{\text{H}^{\text{div}}(\Omega; \mathbb{R}^{2 \times 2})}. \end{aligned}$$

This emphasizes that a conforming approximation of the stress may be advantageous. Furthermore, the discrete inf-sup condition

$$\inf_{\substack{v_h \in U_h \\ \eta_h \in X_h}} \sup_{\tau_h \in \Sigma_{g,h}} \frac{b(\tau_h, v_h) + c(\tau_h, \eta_h)}{\|\tau_h\|_{\text{H}^{\text{div}}} (\|v_h\|_0 + \|\eta_h\|_0)} \geq C > 0 \quad (20)$$

must be fulfilled.

The next theorem ensures the well-posedness of the discrete source problem and provides a "Céa-Lemma" type result. It states that the error of the stress, the displacement, and the skew-symmetric part is quasioptimal, i.e. the errors are up to a constant as good as the best approximations in the corresponding spaces. This constant is independent of the mesh size. Later in the a priori analysis, a separate result for the stress and the skew-symmetric part is needed.

Theorem 3.1. [5, Theorem 3.1] Suppose the spaces $\Sigma_{g,h}$, U_h , and X_h satisfy the inclusion of the kernel property (19) and the inf-sup condition (20). Then (17) admits a unique solution $(u_h, \sigma_h, \gamma_h) \in \Sigma_{g,h} \times U_h \times X_h$, and the following estimate holds:

$$\begin{aligned} & \|\sigma - \sigma_h\|_0 + \|u - u_h\|_0 + \|\gamma - \gamma_h\|_0 \\ & \leq C \left(\inf_{\tau_h \in \Sigma_{g,h}} \|\sigma - \tau_h\|_0 + \inf_{v_h \in U_h} \|u - v_h\|_0 + \inf_{\eta_h \in X_h} \|\gamma - \eta_h\|_0 \right). \end{aligned}$$

In the discussion about the construction of stable finite elements in the context of elasticity with weakly imposed symmetry, we will mainly follow the theory of Boffi, Brezzi, and Fortin [5]. The discrete inf-sup condition

$$\inf_{\substack{v_h \in U_h \\ \eta_h \in X_h}} \sup_{\tau_h \in \Sigma_{g,h}} \frac{(\operatorname{div} \tau_h, v_h) + (\tau_h, \eta_h)}{\|\tau_h\|_{\operatorname{H}^{\operatorname{div}}} (\|v_h\|_0 + \|\eta_h\|_0)} \geq C > 0$$

is valid if there exists a Fortin operator $\Pi_h : \Sigma_g \rightarrow \Sigma_{g,h}$ satisfying

$$\begin{aligned} & (\operatorname{div}(\tau - \Pi_h \tau), v_h) + (\tau - \Pi_h \tau, \eta_h) = 0, \\ & \|\Pi_h \tau\|_{\operatorname{H}^{\operatorname{div}}} \leq C \|\tau\|_{\operatorname{H}^{\operatorname{div}}}, \end{aligned}$$

for any $v_h \in U_h$ and $\eta_h \in X_h$.

Auxiliary Stokes problem

In the construction of stable finite elements in this context, we will use an auxiliary Stokes problem. We assume $\tilde{f} \in H^{-1}(\Omega; \mathbb{R}^2)$ and $\tilde{g} \in L^2_0(\Omega; \mathbb{R})$. The strong formulation of Stokes problem assuming homogeneous Dirichlet boundary conditions for the "velocity" u on the entire boundary states

$$\begin{aligned} -\mu \Delta u - \mu \nabla \operatorname{div} u + \nabla p &= \tilde{f} & \text{in } \Omega, \\ \operatorname{div} u &= \tilde{g} & \text{in } \Omega, \\ u &= 0 & \text{on } \partial\Omega, \end{aligned} \tag{21}$$

where the "pressure" p is only defined up to a constant. The corresponding weak formulation is: find $(u, p) \in H^1_0(\Omega; \mathbb{R}^2) \times L^2(\Omega)/\mathbb{R}$ satisfying

$$\begin{aligned} \mu \int_{\Omega} \nabla(u) : \nabla(v) + \operatorname{div} u \operatorname{div} v \, dx - \int_{\Omega} p \operatorname{div} v \, dx &= \int_{\Omega} \tilde{f} \cdot v \, dx \quad \forall v \in H^1_0(\Omega; \mathbb{R}^2), \\ \int_{\Omega} q \operatorname{div} u \, dx &= \int_{\Omega} \tilde{g} q \, dx \quad \forall q \in L^2(\Omega)/\mathbb{R}. \end{aligned} \tag{22}$$

With the identity

$$\Delta u + \nabla(\operatorname{div} u) = 2 \operatorname{div}(\epsilon(u)) \tag{23}$$

and

$$\epsilon(u) : \nabla v = \epsilon(u) : \epsilon(v)$$

Stokes weak formulation can be reformulated, and we seek for $(u, p) \in H^1_0(\Omega; \mathbb{R}^2) \times L^2(\Omega)/\mathbb{R}$ satisfying

$$\begin{aligned} 2\mu \int_{\Omega} \epsilon(u) : \epsilon(v) \, dx - \int_{\Omega} p \operatorname{div} v \, dx &= \int_{\Omega} \tilde{f} \cdot v \, dx \quad \text{for all } v \in H^1_0(\Omega; \mathbb{R}^2), \\ \int_{\Omega} q \operatorname{div} u \, dx &= \int_{\Omega} \tilde{g} q \, dx \quad \text{for all } q \in L^2(\Omega)/\mathbb{R}. \end{aligned} \tag{24}$$

If $\operatorname{div} u = 0$, the symmetrised gradient ϵ can be replaced by the gradient ∇ by (22). Let us remark, for q smooth:

$$\begin{aligned} \int_{\Omega} \tilde{g} q \, dx &= \int_{\Omega} q \operatorname{div} u \, dx \\ &= - \int_{\Omega} \nabla q \cdot u \, dx + \int_{\partial\Omega} q(u \cdot n) \, ds = - \int_{\Omega} \nabla q \cdot u \, dx. \end{aligned}$$

To restrict ourselves in the second equation to non-constant test functions, the compatibility condition

$$\int_{\Omega} \tilde{g} \, dx = 0$$

is necessary.

Theorem 3.2. [4, Theorem 8.2.1.] *Let $\tilde{f} \in H^{-1}(\Omega; \mathbb{R}^2)$ and $\tilde{g} \in L^2_0(\Omega; \mathbb{R})$ be given. Then there exists a unique solution $(u, p) \in H^1_0(\Omega; \mathbb{R}^2) \times L^2(\Omega)/\mathbb{R}$ satisfying (24) and*

$$\|u\|_1 + \|p\|_0 \leq C(\|f\|_0 + \|g\|_0). \quad (25)$$

Let us return to the construction of stable finite elements in linear elasticity with weakly imposed symmetry. By [5], this can be done in two steps. We assume that the existence of stable finite elements for the stress-displacement formulation is clear. A corresponding theory can be found in [4]. Let us choose $\Sigma_h \times U_h$ such that the following holds: there exists a positive constant C satisfying

$$\inf_{v_h \in U_h} \sup_{\tau_h^1 \in \Sigma_{g,h}} \frac{(\operatorname{div} \tau_h^1, v_h)}{\|v_h\|_0} \geq C \|\tau_h^1\|_{H^{\operatorname{div}}}.$$

Then, for any tensor $\tau \in \Sigma_g$, there exists an interpolation τ_h^1 in $\Sigma_{g,h}$ satisfying

$$\begin{aligned} (\operatorname{div}(\tau - \tau_h^1), v_h) &= 0, \\ \|\tau_h^1\|_{H^{\operatorname{div}}} &\leq \|\tau\|_{H^{\operatorname{div}}}, \end{aligned}$$

for any $v_h \in U_h$. In the second step, we want to construct an interpolation τ_h^2 of τ in $\Sigma_{g,h}$ such that

$$\begin{aligned} (\operatorname{div}(\tau - \tau_h^1 - \tau_h^2), v_h) &= 0, \\ (\tau - \tau_h^1 - \tau_h^2, \gamma_h) &= 0, \\ \|\tau_h^1 + \tau_h^2\|_{H^{\operatorname{div}}} &\leq \|\tau\|_{H^{\operatorname{div}}}, \end{aligned}$$

for any $v_h \in U_h$ and $\gamma_h \in X_h$. If we find such a tensor τ_h^2 , then the full inf-sup condition

$$\inf_{\substack{v_h \in U_h \\ \gamma_h \in X_h}} \sup_{\tau_h^1 + \tau_h^2 \in \Sigma_{g,h}} \frac{(\operatorname{div}(\tau_h^1 + \tau_h^2), v_h)}{\|\gamma_h\|_0 + \|v_h\|_0} \geq C \|\tau_h^1\|_{H^{\operatorname{div}}}$$

is satisfied. In the construction of such a tensor τ_h^2 , we make use of the following Stokes problem. We choose $\tilde{f} = 0$ and $\tilde{g} = \tau_{12} - \tau_{21} + \tau_{h,12}^1 - \tau_{h,21}^1$. Let us assume $V_h \times Q_h$ is a pair of stable finite elements for the Stokes problem satisfying

$$\operatorname{Curl}(V_h) \subset \Sigma_h.$$

Find $(\psi_h, p_h) \in V_h \times Q_h$ such that

$$\begin{aligned} 2\mu \int_{\Omega} \epsilon(\psi_h) : \epsilon(\varphi_h) \, dx - \int_{\Omega} p_h \operatorname{div} \varphi_h \, dx &= 0, \\ \int_{\Omega} q_h \operatorname{div} \psi_h \, dx &= \int_{\Omega} q_h (\tau_{12} - \tau_{21} + \tau_{h,12}^1 - \tau_{h,21}^1) \, dx, \end{aligned} \quad (26)$$

for all $\varphi_h \in V_h$ and $q_h \in Q_h$. With

$$\int_{\Omega} (\tau - \tau_h^1) : S^2(q_h) \, dx = \int_{\Omega} q_h (\tau_{12} - \tau_{21} + \tau_{h,12}^1 - \tau_{h,21}^1) \, dx, \quad (27)$$

the right-hand side of the second equation can be reformulated. Furthermore, the left-hand side of the second equation can be reformulated by

$$\begin{aligned} \int_{\Omega} q_h \operatorname{div} \psi_h \, dx &= \int_{\Omega} q_h \partial_1 \psi_{1,h} + q_h \partial_2 \psi_{2,h} \, dx \\ &= \int_{\Omega} \begin{pmatrix} -\partial_2 \psi_{1,h} & \partial_1 \psi_{1,h} \\ -\partial_2 \psi_{2,h} & \partial_1 \psi_{2,h} \end{pmatrix} : \begin{pmatrix} 0 & q_h \\ -q_h & 0 \end{pmatrix} \, dx \\ &= \int_{\Omega} \operatorname{Curl} \psi_h : S^2(q_h) \, dx, \end{aligned} \quad (28)$$

and the second equation of (26) is

$$\int_{\Omega} \operatorname{Curl} \psi_h : S^2(q_h) \, dx = \int_{\Omega} (\tau - \tau_h^1) : S^2(q_h) \, dx.$$

Setting $\tau_h^2 = \operatorname{Curl} \psi_h$ yields

$$\int_{\Omega} (\tau - \tau_h^1 - \tau_h^2) : S^2(q_h) \, dx = 0 \quad \text{for all } q_h \in Q_h, \quad (29)$$

and τ_h^2 can be identified with a bounded linear operator. The boundedness of τ_h^2 can be seen as follows. The stability estimate (25) yields

$$\|\tau_h^2\|_1 \leq C \|\tau - \tau_h^1\|_0.$$

Let us remark that for any vector v , which is smooth enough, the useful property

$$\operatorname{div}(\operatorname{Curl} v) = \operatorname{div} \begin{pmatrix} -\partial_2 v_1 & \partial_1 v_1 \\ -\partial_2 v_2 & \partial_1 v_2 \end{pmatrix} = \begin{pmatrix} -\partial_1 \partial_2 v_1 + \partial_2 \partial_1 v_1 \\ -\partial_1 \partial_2 v_2 + \partial_2 \partial_1 v_2 \end{pmatrix} = 0 \quad (30)$$

holds. Therefore, we have

$$\operatorname{div} \tau_h^2 = 0.$$

Using the stability of τ_h^1 , we conclude

$$\begin{aligned} \|\tau_h^2\|_{\mathbf{H}^{\operatorname{div}}} &\leq \|\tau - \tau_h^1\|_0 \leq C \|\tau - \tau_h^1\|_{\mathbf{H}^{\operatorname{div}}} \\ &\leq C \left(\|\tau\|_{\mathbf{H}^{\operatorname{div}}} + \|\tau_h^1\|_{\mathbf{H}^{\operatorname{div}}} \right) \\ &\leq C \|\tau\|_{\mathbf{H}^{\operatorname{div}}}. \end{aligned}$$

Finally, stability of τ_h^1 and τ_h^2 yield stability of $\tau_h^1 + \tau_h^2$, i.e.

$$\|\tau_h^1 + \tau_h^2\|_{\mathbf{H}^{\operatorname{div}}} \leq \|\tau_h^1\|_{\mathbf{H}^{\operatorname{div}}} + \|\tau_h^2\|_{\mathbf{H}^{\operatorname{div}}} \leq C \|\tau\|_{\mathbf{H}^{\operatorname{div}}}.$$

Furthermore, and again by (30), we have

$$\int_{\Omega} (\tau - \tau_h^1 - \tau_h^2) : v_h \, dx = 0 \quad \text{for all } v_h \in U_h.$$

This completes the discussion, and the inf-sup condition is valid under certain conditions. According to (29), it is natural to choose X_h as follows:

$$X_h = S^2(Q_h).$$

3.2 Polynomial spaces and finite elements

For $T \subset \mathbb{R}^2$, let the set of polynomials on T with degree less than or equal to k be given by

$$\mathcal{P}_k(T) = \left\{ p : T \rightarrow \mathbb{R} : p(x, y) = \sum_{\substack{i+j \leq k \\ i, j \geq 0}} x^i y^j \right\}.$$

The elements of $\mathcal{P}_k$ are invariant under linear and affine transformations. Further, we define two polynomials on T in a row-vector by

$$\mathcal{P}_k(T; \mathbb{R}^2) = \left\{ p = \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} : p_1, p_2 \in \mathcal{P}_k(T) \right\}.$$

Let us remark that the following interpolation problem is well-posed. On a triangle T with $t = 1 + 2 + \dots + k + 1$ nodes on $k + 1$ parallel lines, for each continuous function f exists a unique polynomial $p \in \mathcal{P}_k(T)$ such that the value of the function f and the value of the polynomial p coincide at each node. Nodes may or may not be on the boundary of the triangle T .

By Ciarlet [14, p. 78], a finite element is a triple (K, P, Σ) for which $\emptyset \neq K \subset \mathbb{R}^d$ is a closed Lipschitz domain, P is a space of real-valued functions over K and Σ is a set of linearly independent functionals over P . In our case, $K = T_{ref} \subset \mathbb{R}^2$ is a reference triangle. The elements in P are called shape functions, and we will restrict ourselves to polynomials. Further, the set Σ defines the degrees of freedom. Every element in P is uniquely determined by the values of the functionals in Σ . In particular, Σ can be specified by a basis of the dual space of P .

Usually, we choose a finite element on a reference triangle and use that polynomials are invariant under linear and affine transformations. The calculations on any element T in the triangulation are done via linear and affine transformations from the reference triangle onto T . Furthermore, a careful elaboration in the error analysis reveals that no order of convergence is lost in this process. For uniform triangulations, the approximation error on the elements in the triangulation is up to a constant as good as the approximation error on the reference element. The constant depends on the uniform triangulation constant κ . These results can be found by various authors and for example, in [7], [8], and [14].

The Lagrange elements are finite elements for which the degrees of freedom are determined by the values at the nodes. The nodal basis functions are functions that have a non-zero value at exactly one node. Let us denote the space of piecewise polynomials, which are globally H^1 by

$$\mathcal{P}_k^c(\mathcal{T}_h) := \{p \in H^1(\Omega) : p|_T \in \mathcal{P}_k(T) \text{ for all } T \in \mathcal{T}_h\},$$

and the space of piecewise polynomials, which are globally L^2 by

$$\mathcal{P}_k^d(\mathcal{T}_h) := \{p \in L^2(\Omega) : p|_T \in \mathcal{P}_k(T) \text{ for all } T \in \mathcal{T}_h\}.$$

For both, $\mathcal{P}_k^c(\mathcal{T}_h)$ and $\mathcal{P}_k^d(\mathcal{T}_h)$, we have

$$\frac{(k+1)(k+2)}{2}$$

degrees of freedom.

It is well-known that any piecewise continuous function is $(n-1)$ times differentiable if and only if it is in $H^n(\Omega)$ on a bounded domain Ω . Therefore, the members of $\mathcal{P}_k^c(\mathcal{T}_h)$ are piecewise continuous polynomials. Unlike the space $\mathcal{P}_k^d$, whose members are generally discontinuous.

Let $P_h^k : L^2(\Omega; \mathbb{R}^2) \rightarrow \mathcal{P}_k^d(\mathcal{T}_h; \mathbb{R}^2)$ be the L^2 -projection onto $\mathcal{P}_k^d(\mathcal{T}_h; \mathbb{R}^2)$. Then locally on each element $T \in \mathcal{T}_h$, we have

$$\int_T (v - P_h^k v) \cdot w = 0 \quad \text{for all } w \in \mathcal{P}_k^d(T; \mathbb{R}^2),$$

and furthermore, the following theorem holds by the famous Bramble-Hilbert-Lemma, and can for example be found in [7] and [8].

Theorem 3.3. *There exists a positive constant C such that*

$$\|v - P_h^k v\|_s \leq Ch^{k+1-s} |v|_{k+1} \quad \text{for all } v \in H^{k+1}(\Omega; \mathbb{R}^2)$$

holds.

The theorem states that the convergence order depends on three things. On the Sobolev norm in which the approximation error is measured, the polynomial degree of the interpolation, and the regularity of the function that is interpolated.

Let us continue by introducing a finite element space for the stress σ . A H^{div} -conforming element is given by the Brezzi-Douglas-Marini BDM_k family. First, we define

$$R_k(\partial T) := \{\phi : \phi \in L^2(\partial T), \phi|_T \in \mathcal{P}_k(E) \text{ for all } E \in \mathcal{E}\},$$

and for any $k \geq 1$, we further define

$$\begin{aligned} \text{BDM}_k(T) &:= \{q \in \mathcal{P}_k(T; \mathbb{R}^2) : p \cdot n \in R_k(\partial T)\}, \\ \text{BDM}_k(\Omega) &:= \{p : p \in \text{BDM}_k(T) \text{ for all } T \in \mathcal{T}_h\}. \end{aligned}$$

The number of degrees of freedom is given by

$$\dim(\text{BDM}_k) = (k+1)(k+2).$$

The $\text{BDM}_k(\Omega)$ -space is an approximation of $H^{\text{div}}(\Omega; \mathbb{R}^2)$. The key is to require the normal component to be continuous across the interelement boundaries. This can be done by defining suitable degrees of freedom. Let us continue to discuss what is meant by an appropriate choice of degrees of freedom. First, we define the following space:

$$\mathbb{H}_k(T) := \{q_k : q_k \in \mathcal{P}_k(T; \mathbb{R}^2), \text{div } q_k = 0 \text{ and } q_k \cdot n = 0 \text{ on } \partial T\}.$$

The set of linear functionals that determine the degrees of freedom is given by a basis of the dual space of $\text{BDM}_k(T)$. The corresponding duality product is the L^2 inner product.

As always for polynomial spaces, the degrees of freedom are given by a basis of $\text{BDM}_k(T)$.

Proposition 3.4. [4, Proposition 2.3.1] *Let $k \geq 1$ and $q \in \text{BDM}_k(T)$. The three conditions*

$$\begin{aligned} \int_{\partial T} q \cdot n \phi \, ds &= 0 \quad \text{for all } \phi \in R_k(\partial T), \\ \int_T q \cdot \nabla p_{k-1} \, dx &= 0 \quad \text{for all } p_{k-1} \in \mathcal{P}_{k-1}(T), \\ \int_T q \cdot p \, dx &= 0 \quad \text{for all } p \in \mathbb{H}_k(T), \end{aligned} \tag{31}$$

imply

$$q = 0.$$

The degrees of freedom can be determined by choosing bases for the spaces $R_k(\partial T)$, $\mathcal{P}_{k-1}(T)$ and $\mathbb{H}_k(T)$. Bases for the first two spaces correspond to the degrees of freedom at the edges, which are values of the normal derivatives. A basis of the third space corresponds to inner degrees of freedom. Inner degrees of freedom appear for $k \geq 2$, while none are in the lowest-order case.

$$\begin{array}{ccc} \mathbb{H}^{\text{div}}(\Omega; \mathbb{R}^{2 \times 2}) & \xrightarrow{\text{div}} & L^2(\Omega; \mathbb{R}^2) \\ \downarrow \Pi_h^k & & \downarrow P_h^{k-1} \\ \Sigma_h & \xrightarrow{\text{div}} & U_h \end{array}$$

Figure 1: Commuting diagram property (33).

Let λ_1 , λ_2 and λ_3 be the barycentric coordinates of the triangle T . Then

$$b_K(x) := \lambda_1(x)\lambda_2(x)\lambda_3(x), \quad x \in T,$$

is a bubble function on T . As it is shown in [4, Lemma 2.3.2], the space $\mathbb{H}_k(T)$ can be characterized by

$$\mathbb{H}_k(T) := \{p_k \in \mathcal{P}_k(T; \mathbb{R}^2) : p_k = \text{Curl}(b_K p_{k-2}), p_{k-2} \in \mathcal{P}_{k-2}(T)\}.$$

This can also be seen by the following: we know $\text{Curl}(b_K p_{k-2}) \in \mathcal{P}_k(T; \mathbb{R}^2)$ and

$$\begin{aligned} \text{div}(\text{Curl}(b_K p_{k-2})) &= 0, \\ \text{Curl}(b_K p_{k-2}) \cdot n &= 0, \end{aligned}$$

for any $p_{k-2} \in \mathcal{P}_{k-2}(T)$.

For $q_1, q_2 \in \text{BDM}_k(\Omega; \mathbb{R}^2)$, let us denote two times row-wise BDM by

$$\begin{pmatrix} q_1 \\ q_2 \end{pmatrix} \in \text{BDM}_k(\Omega; \mathbb{R}^{2 \times 2}).$$

Then let $\Pi^k : \mathbf{H}^{\text{div}}(T; \mathbb{R}^{2 \times 2}) \rightarrow \mathbf{BDM}_k(T; \mathbb{R}^{2 \times 2})$ on each $T \in \mathcal{T}_h$ be defined by the degrees of freedom

$$\begin{aligned} \int_T w_k \cdot (\tau - \Pi^k \tau) \cdot n \, ds &= 0 \quad \text{for all } w_k \in R_k(\partial T; \mathbb{R}^2), \\ \int_T (\tau - \Pi^k \tau) : \nabla p_{k-1} \, dx &= 0 \quad \text{for all } p_{k-1} \in \mathcal{P}_{k-1}(T; \mathbb{R}^2), \\ \int_T (\tau - \Pi^k \tau) : p \, dx &= 0 \quad \text{for all } p \in \mathbb{H}_k(T; \mathbb{R}^{2 \times 2}). \end{aligned} \quad (32)$$

The following Lemma is a direct consequence of [10, Lemma 2.1].

Lemma 3.5. *The operator Π_h^k is uniquely determined by (32). Define the global operator $\Pi_h^k : \mathbf{H}^{\text{div}}(\Omega; \mathbb{R}^{2 \times 2}) \rightarrow \mathbf{BDM}_k(\mathcal{T}_h; \mathbb{R}^{2 \times 2})$ by $\Pi_h^k|_T = \Pi^k(T)$ for any $T \in \mathcal{T}_h$. Then*

$$\text{div } \Pi_h^k = P_h^{k-1} \text{div}, \quad (33)$$

and furthermore

$$\|\tau - \Pi_h^k \tau\|_0 \leq Ch^{k+1} \|\tau\|_{k+1}. \quad (34)$$

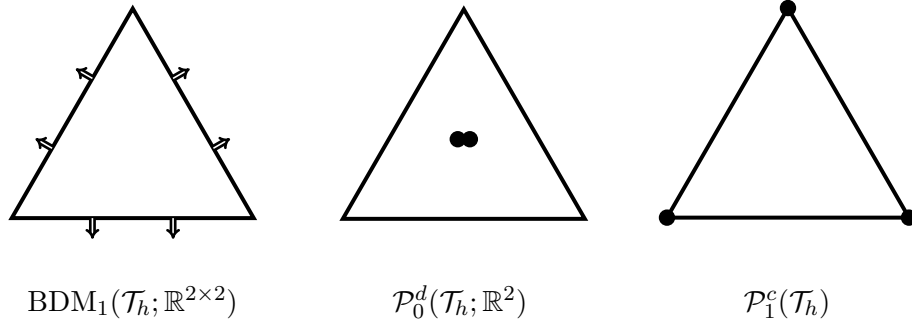


Figure 2: Lowest-order Falk element.

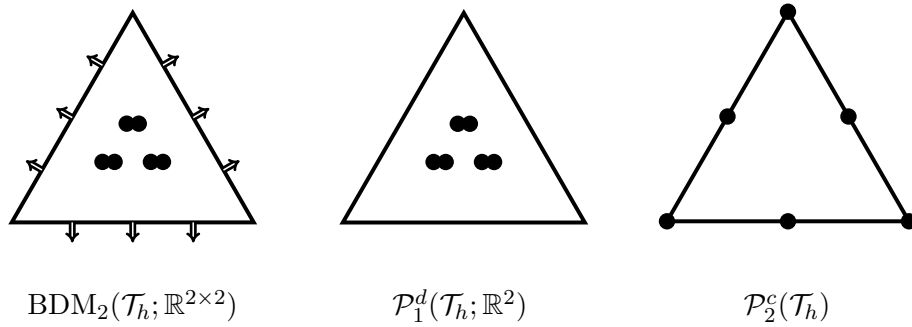


Figure 3: Second-order Falk element.

Remark 3.6. By (32), it is clear that for any $v_h \in \mathcal{P}_k(\Omega; \mathbb{R}^2)$ by choice of the degrees of freedom, the following holds:

$$\begin{aligned}
\int_{\Omega} \operatorname{div}(\tau - \Pi_h^k \tau) \cdot v_h \, dx &= \sum_{T \in \mathcal{T}_h} \int_T \operatorname{div}(\tau - \Pi_h^k \tau) \cdot v_h \, dx \\
&= \sum_{T \in \mathcal{T}_h} - \int_T (\tau - \Pi_h^k \tau) : \nabla v_h \, dx + \int_{\partial T} v \cdot ((\tau - \Pi_h^k \tau) \cdot n) \, ds \\
&= \sum_{T \in \mathcal{T}_h} - \int_T (\tau - \Pi_h^k \tau) : \nabla v_h \, dx + \int_{\partial T} v_h \cdot ((\tau - \Pi_h^k \tau) \cdot n) \, ds \\
&= 0.
\end{aligned}$$

Falk's element

For $k \geq 1$, Falk's element is given by

$$\begin{aligned}
\Sigma_{g,h} &:= \operatorname{BDM}_k(\mathcal{T}_h; \mathbb{R}^{2 \times 2}), \\
U_h &:= \mathcal{P}_{k-1}^d(\mathcal{T}_h; \mathbb{R}^2), \\
X_h &:= \operatorname{S}^2(\mathcal{P}_k^c(\mathcal{T}_h)).
\end{aligned}$$

It has been previously discussed that the stability of an element in the context of elasticity with reduced symmetry is closely related to an element that is stable for the Stokes problem. The generalized Taylor-Hood element is a corresponding Stokes stable element and for $k \geq 1$, this is

$$V_h \times Q_h := \mathcal{P}_{k+1}^c(\mathcal{T}_h; \mathbb{R}^2) \times \mathcal{P}_k^c(\mathcal{T}_h).$$

Both the "velocity" and the "pressure" are continuously approximated. For Falk's element, a corresponding weakly elasticity complex is given by Figure 4. A similar complex can be found in [21]. Let $\hat{Q}_h$ be the L^2 -projection onto Q_h . By (28), the equation

$$\operatorname{S}^2 \hat{Q}_h \operatorname{div} \psi_h = 2 \operatorname{S}^2 \hat{Q}_h \operatorname{as} \operatorname{Curl} \psi_h$$

holds and makes the diagram commute.

$$\begin{array}{ccccc}
& & X_h = \operatorname{S}^2(Q_h) & & \\
& \nearrow & & \nwarrow & \\
& \operatorname{S}^2 \hat{Q}_h \operatorname{div} & & 2 \operatorname{S}^2 \hat{Q}_h \operatorname{as} & \\
V_h & \xrightarrow{\operatorname{Curl}} & \Sigma_{g,h} & \xrightarrow{\operatorname{div}} & U_h
\end{array}$$

Figure 4: A Commuting diagram property in elasticity with reduced symmetry.

Remark 3.7. Let $I_h^k : L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}) \rightarrow X_h = S^2(\mathcal{P}_k^c(\mathcal{T}_h))$ be an interpolation operator. For the postprocessing, we need a result similar to Theorem 3.1 where the approximation error $\|\sigma - \sigma_h\|_0 + \|\gamma - \gamma_h\|_0$ is decoupled from the approximation error $\|u - u_h\|_0$. A corresponding statement can be found in Lee [21, Theorem 2],

$$\|\sigma - \sigma_h\|_0 + \|\gamma - \gamma_h\|_0 \leq C \left(\|\sigma - \Pi_h^k \sigma\|_0 + \|\gamma - I_h^k \gamma\|_0 \right). \quad (35)$$

To simplify the notation, we write $P_h := P_h^{k-1}$ in the following.

4 A priori error analysis

In this section, we aim to analyse the eigenvalue problem in elasticity with weakly imposed symmetry. To this end, some spectral theoretical definitions and properties are discussed first. It is well-known that the basic reference for the treatment of finite element approximation of compact eigenvalue problems is the Babuška-Osborn theory, which was originally introduced in [2].

This analysis cannot simply be applied to mixed problems because the corresponding solution operator is generally not compact. A framework to deal with mixed eigenvalue problems is given by Mercier et al. [22] and by Boffi [3]. It turns out that the existence of a Fortin operator is crucial, which is related to the inf-sup condition but slightly stronger.

According to the eigenvalue theory for mixed problems, it can be shown that the discrete eigenvalues converge to the continuous eigenvalues, the discrete eigenspaces converge to the continuous eigenspaces, and that no spurious eigenvalues pollute the spectrum.

Furthermore, an identity between the discrete and the continuous eigenvalue is derived, and a corresponding a priori Falk finite element approximation error is presented.

For the postprocessing, an estimate between the L^2 -projection of u and the discrete solution u_h is needed. Here, we need to pay particular attention to the lowest-order Falk element.

4.1 The eigenvalue problem

We start by introducing some notations that correspond to spectral theory. On a Hilbert space H , let $T : H \rightarrow H$ be a linear, well-defined and compact operator. The resolvent of T is given by

$$\rho(T) = \{\kappa \in \mathbb{C} : (\kappa I - T) \text{ is bijective} \},$$

and typically, the spectrum of T is defined as

$$\sigma(T) := \mathbb{C} / \rho(T).$$

In the context of compact eigenvalue problems on Hilbert spaces, the spectrum consists almost only of eigenvalues. Every $\kappa \in \sigma(T) / \{0\}$ is an eigenvalue of T with finite multiplicity, i.e. for every eigenvalue κ the associated eigenspace $\ker(\kappa I - T)$ is finite dimensional and the spectrum $\sigma(T)$ does not have a limit point away from zero.

For any $\kappa \in \mathbb{R}$, the algebraic multiplicity, which is always greater or equal to one, is given by the dimension of $\ker(\kappa I - T)^\alpha$ and the geometric multiplicity is given by the dimension of the eigenspace $\ker(\kappa I - T)$. The geometric multiplicity is always less or equal to the algebraic multiplicity. If T is self-adjoint, then algebraic multiplicity and geometric multiplicity coincide.

The elements of $\ker(\kappa I - T)$ are called the eigenfunctions of T associated with the eigenvalue κ and we order the eigenvalues so that

$$\kappa_1 \leq \kappa_2 \leq \dots \leq \kappa_j \leq \dots$$

The values of several κ_j can coincide according to the multiplicity of the associated eigenvalue.

The mixed eigenvalue problem in elasticity with weakly imposed symmetry

We set $f = -\kappa u$ in the variational formulation of the problem (11). For simplicity, we assume $\kappa \in \mathbb{R}$ is a simple eigenvalue, i.e. the algebraic multiplicity equals to one. The corresponding eigenspace is then one-dimensional. It should be remarked that even for simple eigenvalues the eigenfunctions are not uniquely defined. Only up to their sign.

The eigenfunctions have the physical meaning of displacements, and for a given eigenvalue, we seek for $\kappa \in \mathbb{R}$ and $u \in U$ with $u \neq 0$ such that, for some $(\sigma, \gamma) \in \Sigma_g \times X$,

$$\begin{aligned} (\mathbb{C}^{-1}\sigma, \tau) + (\operatorname{div} \tau, u) + (\gamma, \tau) &= 0 & \text{for all } \tau \in \Sigma_0, \\ (\operatorname{div} \sigma, v) &= -\kappa(u, v) & \text{for all } v \in U, \\ (\sigma, \eta) &= 0 & \text{for all } \eta \in X. \end{aligned} \quad (36)$$

In the discrete eigenvalue problem, we seek for $\kappa_h \in \mathbb{R}$ and $u \in U_h$ with $u_h \neq 0$ satisfying, for some $(\sigma_h, \gamma_h) \in \Sigma_{g,h} \times X_h$,

$$\begin{aligned} (\mathbb{C}^{-1}\sigma_h, \tau_h) + (\operatorname{div} \tau_h, u_h) + (\gamma_h, \tau_h) &= 0 & \text{for all } \tau_h \in \Sigma_{0,h}, \\ (\operatorname{div} \sigma_h, v_h) &= -\kappa_h(u_h, v_h) & \text{for all } v_h \in U_h, \\ (\sigma_h, \eta_h) &= 0 & \text{for all } \eta_h \in X_h. \end{aligned} \quad (37)$$

Both (36) and (37) are well-posed problems by the discussion of sections 2 and 3.

Following the theory presented in [3, Section 14], we are in the case of symmetric eigenvalue problems since $a(\bullet, \bullet)$ is symmetric. The spaces Σ_g , U , and X are Hilbert spaces, and we define the operators $R : U \rightarrow \Sigma_g$, $S : U \rightarrow U$ and $T : U \rightarrow X$ by

$$Rf = \sigma^f, \quad Sf = u^f, \quad Tf = \gamma^f.$$

These operators are linear and well-defined by the well-posedness of the source problem. For given $f \in U$, we seek for $(Rf, Sf, Tf) \in \Sigma_g \times U \times X$ satisfying

$$\begin{aligned} (\mathbb{C}^{-1}Rf, \tau) + (\operatorname{div} \tau, Sf) + (Tf, \tau) &= 0 & \text{for all } \tau \in \Sigma_0, \\ (\operatorname{div} Rf, v) &= -(f, v) & \text{for all } v \in U, \\ (Rf, \eta) &= 0 & \text{for all } \eta \in X. \end{aligned}$$

Further, let us introduce discrete operators $R_h : U \rightarrow \Sigma_{g,h}$, $S_h : U \rightarrow U_h$, and $T_h : U \rightarrow X_h$. Find $(R_h f, S_h f, T_h f) \in \Sigma_{g,h} \times U_h \times X_h$ satisfying

$$\begin{aligned} (\mathbb{C}^{-1}R_h f, \tau_h) + (\operatorname{div} \tau_h, S_h f) + (T_h f, \tau_h) &= 0 & \text{for all } \tau_h \in \Sigma_{0,h}, \\ (\operatorname{div} R_h f, v_h) &= -(f, v_h) & \text{for all } v_h \in U_h, \\ (R_h f, \eta_h) &= 0 & \text{for all } \eta_h \in X_h. \end{aligned} \quad (38)$$

Both, the continuous and the discrete problem are well-posed by the theory of sections 2 and 3. Since $U_h \subset U$, let us remark that the following orthogonality condition

$$(\operatorname{div} Rf, v_h) - (\operatorname{div} R_h f, v_h) = (f - f, v_h) = 0 \quad (39)$$

holds.

To make statements regarding the convergence of eigenvalues and eigenfunctions, we need the definition of the gap between Hilbert spaces.

Definition 4.1. [3, p. 49] Let H_1 and H_2 be Hilbert spaces. Then, we define

$$\delta(H_1, H_2) = \sup_{u \in H_1, \|u\|_{H_1}=1} \inf_{v \in H_2} \|u - v\|_{H_2},$$

and, furthermore, the gap between H_1 and H_2 is given by

$$\hat{\delta}(H_1, H_2) = \max(\delta(H_1, H_2), \delta(H_2, H_1)).$$

Definition 4.2. [3, p. 49] We say the discrete eigenvalue problem converges to the continuous one if for every $\epsilon > 0$ and $k > 0$, there is some $h_0 > 0$ such that for any $h < h_0$, we have

$$\max_{1 \leq i \leq m(k)} |\kappa^{(i)} - \kappa_h^{(i)}| \leq \epsilon,$$

$$\hat{\delta}(U, U_h) \leq \epsilon.$$

Proposition 4.3. [3, Proposition 7.4] The convergence defined in Definition 4.2 is valid if and only if

$$\|R - R_h\|_{\mathcal{L}} \rightarrow 0, \quad \|S - S_h\|_{\mathcal{L}} \rightarrow 0, \quad \|T - T_h\|_{\mathcal{L}} \rightarrow 0.$$

We therefore need to show uniform convergence of the solution operators. Here, we will continue to follow the theory of Boffi [3, Section 14] with some necessary arguments for our problem.

Let $\Sigma_g^0 \subset H^{\text{div}}(\Omega; \mathbb{R}^{2 \times 2})$ and $U^0 \subset L^2(\Omega; \mathbb{R}^2)$, $X^0 \subset L_{\text{skw}}(\Omega; \mathbb{R}^{2 \times 2})$ contain all solutions (σ, u, γ) satisfying

$$Rf = \sigma, \quad Sf = u, \quad Tf = \gamma,$$

as f varies in $L^2(\Omega; \mathbb{R}^2)$. These spaces contain functions with at least slightly better regularity, and this is for $\epsilon > 0$,

$$\Sigma^0 \subset H^\epsilon(\Omega; \mathbb{R}^{2 \times 2}),$$

$$U^0 \subset H^\epsilon(\Omega; \mathbb{R}^2),$$

$$X^0 \subset H^\epsilon(\Omega; \mathbb{R}^{2 \times 2}),$$

depending on the domain Ω .

Definition 4.4. [3, Definition 14.1] (Weak approximability)

We call U_0 weakly approximable w.r.t. the weighted L^2 -inner product $\|\bullet\|_a = (\mathbb{C}^{-1}\bullet, \bullet)$ if there exists a constant $C_W(h)$, tending to zero as h goes to zero, such that

$$(\text{div } \tau_h, v) \leq C_W(h) \|\tau_h\|_a \|v\|_{U^0}$$

for any $\tau_h \in \ker(B_h + C_h)$ and $v \in U^0$.

Definition 4.5. [3, Definition 14.2] (Strong approximability)

We call U^0 strongly approximable if there exists a constant $C_S(h)$, tending to zero as h goes to zero, such that for any $v \in U^0$, there is some $v_h \in U_h$ satisfying

$$\|v - v_h\|_0 \leq C_S(h) \|v\|_{U^0}.$$

Definition 4.6. [3, Definition 14.5](Fortid condition)

We say the Fortid condition is satisfied if there is a Fortin operator $\tilde{\Pi}_h$ and furthermore there is a constant $C_F(h)$, tending to zero as h goes to zero, such that

$$\|\sigma - \tilde{\Pi}_h \sigma\|_0 \leq C_F(h) \|\sigma\|_{\Sigma_g^0}.$$

Theorem 4.7. [3, Theorem 14.6] If the weak approximability of U^0 w.r.t. $(\bullet, \bullet)_a$, the strong approximability of U^0 and the Fortid condition are satisfied, then exists a constant $C(h)$ with $C(h) \rightarrow 0$ as $h \rightarrow 0$ such that

$$\|Rf - R_h f\|_0 + \|Sf - S_h f\|_0 + \|Tf - T_h f\|_0 \leq C(h) \|f\|_0$$

for any $f \in L^2(\Omega; \mathbb{R}^{2 \times 2})$.

Let us continue verifying the weak approximability of U^0 w.r.t. $(\bullet, \bullet)_a$ and the strong approximability of U^0 . The strong approximability is a consequence of higher regularity in U^0 . For any $v \in U^0$, we have

$$\|v - P_h v\|_0 \leq Ch^\epsilon \|v\|_{H^\epsilon} \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

To show the weak approximability, for $v \in U^0$ and $\tau_h \in \ker(B_h + C_h)$, we have

$$(\operatorname{div} \tau_h, v) = (\operatorname{div} \tau_h, v - P_h v)$$

since τ_h is in the kernel of $B_h + C_h$ and $P_h v \in U_h$. Boundedness of $b(\bullet, \bullet)$ yields

$$(\operatorname{div} \tau_h, v - P_h v) \leq \|\tau_h\|_{H^{\operatorname{div}}} \|v - P_h v\|_0.$$

Since $\operatorname{div}(\Sigma_{g,h}) \subset U_h$, we have $\|\operatorname{div} \tau_h\|_0 = 0$ for any $\tau_h \in \ker(B_h + C_h)$. Then

$$\|\tau_h\|_{H^{\operatorname{div}}} \|v - P_h v\|_0 = \|\tau_h\|_0 \|v - P_h v\|_0.$$

Finally, strong approximability concludes

$$\begin{aligned} (\operatorname{div} \tau_h, v) &\leq \|\tau_h\|_0 \|v - P_h v\|_0 \\ &\leq Ch^\epsilon \|\tau_h\|_a \|v\|_{H^\epsilon} \rightarrow 0 \quad \text{as } h \rightarrow 0. \end{aligned}$$

We continue to show the Fortid condition. For any $\tau \in \Sigma_g$, the existence of a Fortin operator $\tilde{\Pi}_h \tau := \tau_h^1 + \tau_h^2$ for the Falk element has been shown in section 3 and we have

$$\begin{aligned} (\operatorname{div}(\tau - \tau_h^1 - \tau_h^2), v_h) + (\tau - \tau_h^1 - \tau_h^2, \eta_h) &= 0, \\ \|\tau_h^1 + \tau_h^2\|_0 &\leq C \|\tau\|_0, \end{aligned}$$

for any $v_h \in U_h$ and $\eta \in X_h$. Moreover, we have

$$\|\tau - \tilde{\Pi}_h \tau\|_0 \leq Ch^\epsilon \|\tau\|_\epsilon.$$

All assumptions of Theorem 4.7 have been verified and we conclude

$$\begin{aligned} \|R - R_h\|_{\mathcal{L}(U, \Sigma)} &\rightarrow 0 \quad \text{as } h \rightarrow 0, \\ \|S - S_h\|_{\mathcal{L}(U, U)} &\rightarrow 0 \quad \text{as } h \rightarrow 0, \\ \|T - T_h\|_{\mathcal{L}(U, X)} &\rightarrow 0 \quad \text{as } h \rightarrow 0. \end{aligned}$$

By Proposition 4.3, we have

$$\hat{\delta}(U, U_h) \leq C(\|R - R_h\|_{\mathcal{L}(U, \Sigma)} + \|S - S_h\|_{\mathcal{L}(U, U)} + \|T - T_h\|_{\mathcal{L}(U, X)})$$

so that

$$\hat{\delta}(U, U_h) \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

Let us return briefly to the source problem. The following theorem presents estimates between the continuous and the discrete solution of the source problem. These are needed to derive the a priori Falk finite element eigenvalue approximation.

Theorem 4.8. *Suppose the solution (u, σ, γ) of the source problem (11) admits higher regularity, i.e.*

$$u \in H^k(\Omega; \mathbb{R}^2), \quad \sigma \in H^{k+1}(\Omega; \mathbb{R}^{2 \times 2}), \quad \gamma \in H^{k+1}(\Omega; \mathbb{R}^{2 \times 2}) \cap L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}),$$

and suppose $(u_h, \sigma_h, \gamma_h)$ is the solution of the discrete boundary eigenvalue problem (17). Then the following estimates hold:

$$\begin{aligned} \|u - u_h\|_0 &\leq Ch^k \left(|u|_k + |\sigma|_k + |\gamma|_k \right), \\ \|\sigma - \sigma_h\|_0 + \|\gamma - \gamma_h\|_0 &\leq Ch^{k+1} \left(|\sigma|_{k+1} + |\gamma|_{k+1} \right). \end{aligned}$$

Proof. The first inequality follows by Theorem 3.1 and (34). The second inequality follows by (35) and (34). $\square$

The previous Theorem yields the following convergence rates for of the solution operators:

$$\begin{aligned} \|Rf - R_h f\|_0 + \|Tf - T_h f\|_0 &\leq Ch^{k+1} (|\sigma^f|_{k+1} + |\gamma^f|_{k+1}), \\ \|Sf - S_h f\|_0 &\leq Ch^k (|u^f|_k + |\sigma^f|_k + |\gamma^f|_k). \end{aligned} \tag{40}$$

In the following Lemma, we derive a useful identity for the difference between the continuous and the discrete eigenvalue. It is interesting to note that the continuous solution, provided it is smooth enough, satisfies

$$\mathbb{C}^{-1}\sigma + \gamma = \nabla u. \tag{41}$$

This does not hold for the discrete solution to our problem (37), but later in the postprocessing, we will construct the postprocessed eigenfunction using this idea.

Due to the following scaling argument, we can assume the solution u to be normalized. Suppose $\kappa \in \mathbb{R}$ and $u \in U$ solve, for some $(\sigma, \gamma) \in \Sigma_g \times X$, the eigenvalue problem (36), then

$$\begin{aligned} \left(\mathbb{C}^{-1} \left(\frac{\sigma}{\|u\|_0} \right), \tau \right) + \left(\operatorname{div} \tau, \frac{u}{\|u\|_0} \right) + \left(\frac{\gamma}{\|u\|_0}, \tau \right) &= 0 & \text{for any } \tau \in \Sigma_g, \\ \left(\operatorname{div} \frac{\sigma}{\|u\|_0}, v \right) &= -\kappa \left(\frac{u}{\|u\|_0}, v \right) & \text{for any } v \in U, \\ \left(\frac{\sigma}{\|u\|_0}, \eta \right) &= 0 & \text{for any } \eta \in X. \end{aligned}$$

Therefore, $\kappa \in \mathbb{R}$ and $\frac{u}{\|u\|_0} \in U$ solve, for some $(\sigma, \gamma) \in \Sigma_g \times X$, the eigenvalue problem (36). Analogously, one can show that the discrete solution u_h can be assumed to be normalized.

Lemma 4.9. *The following identity*

$$\kappa - \kappa_h = \|\mathbb{C}^{-1}(\sigma - \sigma_h) + (\gamma - \gamma_h)\|_{\mathbb{C}^{\frac{1}{2}}}^2 - 2\mu\|\gamma - \gamma_h\|_0^2 - \kappa_h\|u - u_h\|_0^2$$

holds.

Proof. We assume u, u_h are normalized, i.e.

$$(u, u) = 1 \quad \text{and} \quad (u_h, u_h) = 1.$$

Then by (36) and (37), it follows

$$\begin{aligned} \|\mathbb{C}^{-1/2}(\sigma - \sigma_h)\|_0^2 &= (\mathbb{C}^{-1}\sigma, \sigma) + (\mathbb{C}^{-1}\sigma_h, \sigma_h) - 2(\mathbb{C}^{-1}\sigma, \sigma_h) \\ &= -(\operatorname{div} \sigma, u) - (\gamma, \sigma) - (\operatorname{div} \sigma_h, u_h) - (\gamma_h, \sigma_h) + 2(\operatorname{div} \sigma_h, u) \\ &\quad + 2(\gamma, \sigma_h) \\ &= \kappa + \kappa_h + 2(\operatorname{div} \sigma_h, u) + 2(\gamma, \sigma_h), \end{aligned}$$

since by the third equation of (36) and (37), we have

$$\begin{aligned} (\eta, \sigma) &= 0 \quad \text{for all } \eta \in X, \\ (\eta_h, \sigma_h) &= 0 \quad \text{for all } \eta_h \in X_h. \end{aligned}$$

The equation

$$\operatorname{div} \sigma_h + \kappa_h u_h = 0$$

holds strongly. Multiplying with $u \in L^2(\Omega; \mathbb{R}^2)$ and integrating over Ω yields

$$(\operatorname{div} \sigma_h, u) = -\kappa_h(u_h, u).$$

Then we have

$$\begin{aligned} \|\mathbb{C}^{-\frac{1}{2}}(\sigma - \sigma_h)\|_0^2 &= \kappa - \kappa_h + \kappa_h(2 - 2(u_h, u)) - 2(\gamma - \gamma_h, \sigma - \sigma_h) \\ &= \kappa - \kappa_h + \kappa_h\|u - u_h\|_0^2 - 2(\gamma - \gamma_h, \sigma - \sigma_h), \end{aligned}$$

and the last term can be rewritten to

$$\begin{aligned} 2(\gamma - \gamma_h, \sigma - \sigma_h) &= 2(\mathbb{C}^{\frac{1}{2}}(\gamma - \gamma_h), \mathbb{C}^{-\frac{1}{2}}(\sigma - \sigma_h)) \\ &= \|\mathbb{C}^{-1}(\sigma - \sigma_h) + (\gamma - \gamma_h)\|_{\mathbb{C}^{\frac{1}{2}}}^2 - \|\mathbb{C}^{-\frac{1}{2}}(\sigma - \sigma_h)\|_0^2 \\ &\quad - \|\mathbb{C}^{\frac{1}{2}}(\gamma - \gamma_h)\|_0^2. \end{aligned}$$

Skew-symmetric matrices are trace free, which yields $(\mathbb{C}\gamma, \sigma) = 2\mu(\gamma, \sigma)$, and we conclude

$$\kappa - \kappa_h = \|\mathbb{C}^{-1}(\sigma - \sigma_h) + (\gamma - \gamma_h)\|_{\mathbb{C}^{\frac{1}{2}}}^2 - 2\mu\|\gamma - \gamma_h\|_0^2 - \kappa_h\|u - u_h\|_0^2.$$

□

Theorem 4.10. *Assume $u \in H^k(\Omega; \mathbb{R}^2)$, $\sigma \in H^k(\Omega; \mathbb{R}^{2 \times 2})$ and $\gamma \in H^k(\Omega; \mathbb{R}^{2 \times 2}) \cap L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2})$. Then the following estimate for the Falk finite element eigenvalue approximation*

$$|\kappa - \kappa_h| \leq Ch^{2k}(\|\sigma\|_k + \|u\|_k + \|\gamma\|_k)$$

holds.

Proof. The proof is a consequence of Theorem 4.8 and Lemma 4.9. □

The eigenvalue identity and the previous theorem reveal that the eigenvalue approximation is as good as the eigenfunction approximation squared. In particular, this is the restricting term. Improving the eigenfunction convergence through a post-processing also improves the eigenvalue approximation error.

4.2 An estimate for $\|P_h u - u_h\|_0$

For the L^2 -projection P_h onto U_h , it holds

$$(P_h v)(v_h) = (v, v_h) \quad \text{for all } v_h \in U_h.$$

We need for our postprocessing an estimate of the form

$$\|P_h u - u\|_0 \leq Ch^{k+2}(|\sigma|_{k+1} + |\gamma|_{k+1}).$$

The source problem: an estimate for $\|P_h u - u_h\|_0$

First, let us assume $f \in U_h$ and derive such an estimate for the source problem. It is clear that $f = -\kappa u \notin U_h$ for the eigenvalue problem in general. Later, using a triangle inequality, we derive an estimate of the form (4.2) for the eigenvalue problem. For a given $f \in U_h$, let $(\sigma, u, \gamma) \in \Sigma_g \times U \times X$ be the solution to the source problem (11) and let $(\sigma_h, u_h, \gamma_h) \in \Sigma_{g,h} \times U_h \times X_h$ be the solution to the corresponding discrete problem (17).

Moreover, let us assume the solution $\sigma \in H^1(\Omega; \mathbb{R}^{2 \times 2})$, $u \in H^2(\Omega; \mathbb{R}^2)$ and $\gamma \in H^1(\Omega; \mathbb{R}^{2 \times 2}) \cap L^2_{\text{skw}}(\Omega; \mathbb{R}^{2 \times 2})$ admits some additional regularity and that the regularity estimate

$$\|\sigma\|_1 + \|u\|_2 + \|\gamma\|_1 \leq C\|f\|_0 \quad (42)$$

holds. Then, a duality argument in the sense of Aubin-Nitsche can be applied to get an additional h -power. It may be remarked that the regularity of the solution depends on the domain and may not be given in practice.

Lemma 4.11. *For given $f \in U_h$ and $k \geq 2$, let $\sigma \in H^{k+1}(\Omega; \mathbb{R}^{2 \times 2})$ and $\gamma \in H^{k+1}(\Omega; \mathbb{R}^{2 \times 2}) \cap L^2_{\text{skw}}(\Omega; \mathbb{R}^{2 \times 2})$. Then the estimate*

$$\|P_h u - u_h\|_0 \leq Ch^{k+2}(|\sigma|_{k+1} + |\gamma|_{k+1})$$

holds.

Proof. Let

$$(\pi, z, \mu) \in \Sigma_g \cap H^1(\Omega; \mathbb{R}^{2 \times 2}) \times U \cap H^2(\Omega; \mathbb{R}^2) \times X \cap H^1(\Omega; \mathbb{R}^{2 \times 2}) \cap L^2_{\text{skw}}(\Omega; \mathbb{R}^{2 \times 2})$$

be the solution to the auxiliary problem

$$\begin{aligned} (\mathbb{C}^{-1}\pi, q) + (\text{div } q, z) + (q, \mu) &= 0 & \text{for all } q \in \Sigma_{0,h}, \\ (\text{div } \pi, p) &= -(u_h - P_h u, p) & \text{for all } p \in U, \\ (\pi, o) &= 0 & \text{for all } o \in X. \end{aligned} \quad (43)$$

We know that such a solution exists and is unique due to the discussion in Chapter 2. By (42), the solution (π, z, μ) satisfies the regularity estimate

$$\|\pi\|_1 + \|z\|_2 + \|\mu\|_1 \leq C\|u_h - P_h u\|_0.$$

Let $(\pi_h, z_h, \mu_h) \in \Sigma_{g,h} \times U_h \times X_h$ be the solution to the corresponding discrete problem

$$\begin{aligned} (\mathbb{C}^{-1}\pi_h, q_h) + (\text{div } q_h, z_h) + (q_h, \mu_h) &= 0 & \text{for all } q_h \in \Sigma_{0,h}, \\ (\text{div } \pi_h, p_h) &= -(u_h - P_h u, p_h) & \text{for all } p_h \in U, \\ (\pi_h, o_h) &= 0 & \text{for all } o_h \in X_h. \end{aligned} \quad (44)$$

Further, let $\Pi_h \pi$, $P_h z$, and $\tilde{\mu}$ be the orthonormal projections of π onto $\Sigma_{g,h}$, z onto U_h , and μ onto X_h . For fixed the test functions in (43)

$$q = \sigma - \sigma_h \in \Sigma_{g,h}, \quad p = u_h - P_h u \in U_h, \quad o = \gamma - \gamma_h \in X_h,$$

we obtain

$$\begin{aligned} \|u_h - P_h u\|_0^2 &= (\mathbb{C}^{-1} \pi, \sigma - \sigma_h) + (\operatorname{div}(\sigma - \sigma_h), z) + (\sigma - \sigma_h, \mu) \\ &\quad - (\operatorname{div} \pi, u_h - P_h u) + (\pi, \gamma - \gamma_h). \end{aligned}$$

For fixed test functions $\tau = \Pi_h \pi$, $v = \tilde{z}$, $\eta = \tilde{\mu}$ in the mixed problem (11) and the corresponding discrete mixed problem (17), with the new problem (43), and (6), we have the following:

$$\begin{aligned} \|u_h - P_h u\|_0^2 &= (\mathbb{C}^{-1} \pi, \sigma - \sigma_h) + (\operatorname{div}(\sigma - \sigma_h), z) + (\sigma - \sigma_h, \mu) \\ &\quad - (\operatorname{div} \pi, u_h - P_h u) + (\pi, \gamma - \gamma_h) \\ &\quad - (\mathbb{C}^{-1}(\sigma - \sigma_h), \Pi_h \pi) - (\operatorname{div} \Pi_h \pi, u - u_h) - (\Pi_h \pi, \gamma - \gamma_h) \\ &\quad - (\operatorname{div}(\sigma - \sigma_h), P_h z) - (\sigma - \sigma_h, \tilde{\mu}) \\ &= (\mathbb{C}^{-1}(\sigma - \sigma_h), \pi - \Pi_h \pi) + (\operatorname{div}(\sigma - \sigma_h), z - P_h z) + (\sigma - \sigma_h, \mu - \tilde{\mu}) \\ &\quad + (\pi - \Pi_h \pi, \gamma - \gamma_h) + (\operatorname{div}(\pi - \Pi_h \pi), P_h u - u_h) \\ &\quad + (\operatorname{div} \Pi_h \pi, P_h u - u). \end{aligned} \tag{45}$$

In the construction of stable finite elements, e.g. Section 3, we have seen that the interpolation operator Π_h is a bounded Fortin operator. Since $u_h - P_h u \in U_h$, we have

$$(\operatorname{div}(\pi - \Pi_h \pi), u_h - P_h u) = 0.$$

Furthermore, by

$$\operatorname{div}(\Sigma_{g,h}) \subset U_h,$$

we have $\operatorname{div} \Pi_h \pi \in U_h$ and for the projection operator P_h , it holds

$$(\operatorname{div} \Pi_h \pi, P_h u - u) = 0.$$

Therefore, the last two terms in (45) vanish. The equilibrium condition yields

$$(\operatorname{div}(\sigma - \sigma_h), P_h z - z) = 0.$$

Using (42), we finally conclude

$$\begin{aligned} \|u_h - P_h u\|_0^2 &\leq C \|\pi - \Pi_h \pi\|_0 \|\sigma - \sigma_h\|_0 + \|\sigma - \sigma_h\|_0 \|\mu - \tilde{\mu}\|_0 \\ &\quad + \|\pi - \Pi_h \pi\|_0 \|\gamma - \gamma_h\|_0 \\ &\leq Ch \left(\|\sigma - \sigma_h\|_0 (|\pi|_1 + |\mu|_1) + \|\gamma - \gamma_h\|_0 |\pi|_1 \right) \\ &\leq Ch (\|\sigma - \sigma_h\|_0 + \|\gamma - \gamma_h\|_0) \|u_h - P_h u\|_0 \\ &\leq Ch^{k+2} (|\sigma|_{k+1} + |\gamma|_{k+1}) \|u_h - P_h u\|_0. \end{aligned}$$

□

Remark 4.12. If $f \in U$ and $k \geq 2$, then

$$\begin{aligned} 0 \neq (\operatorname{div}(\sigma - \sigma_h), P_h z - z) &\leq \|\operatorname{div}(\sigma - \sigma_h)\|_0 \|P_h z - z\|_0 \\ &\leq Ch^{k+2} |\sigma|_{k+1} |z|_2 \\ &\leq Ch^{k+2} |\sigma|_{k+1} \|u_h - P_h u\|_0. \end{aligned}$$

Therefore, the statement of Lemma 4.2 is also valid for $f \in U$ if the lowest-order case is excluded.

Remark 4.13. In the lowest-order case, $k = 1$, and without the assumption $f \in U_h$, we only have

$$\|P_h u - u_h\|_0 \leq Ch^2(|\sigma|_2 + |\gamma|_2).$$

This is also the result one gets in the lowest-order case, using the Brezzi-Douglas-Marini family for the mixed Poisson problem if $f \notin U_h$, c.f. [25, Theorem 2.1].

The eigenvalue problem: an estimate for $\|P_h u - u_h\|_0$

For the eigenvalue problem, we have $f = \kappa u \notin U_h$ and therefore generally we do not have an orthogonality condition like

$$(\operatorname{div}(\sigma - \sigma_h), v_h) = (\kappa u - \kappa_h u_h, v_h) \neq 0.$$

But (39) yields

$$(\operatorname{div}(\sigma - R_h(\kappa u)), v_h) = (\kappa u - \kappa_h u, v_h) = 0 \quad (46)$$

for all $v_h \in U_h$. Applying a triangle inequality, we can estimate $\|P_h u - u_h\|_0$ in two steps:

$$\|P_h u - u_h\|_0 \leq \|P_h u - S_h(\kappa u)\|_0 + \|S_h(\kappa u) - u_h\|_0. \quad (47)$$

An estimate for the first term in (47) follows by Lemma 4.11. The second term is estimated in the following lemma.

Lemma 4.14. For $k \geq 2$, assume $\sigma \in H^{k+1}(\Omega; \mathbb{R}^{2 \times 2})$ and $\gamma \in H^{k+1}(\Omega; \mathbb{R}^{2 \times 2}) \cap L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2})$. Then

$$\|S_h(\kappa u) - u_h\|_0 \leq Ch^{k+2}(|\sigma|_{k+1} + |\gamma|_{k+1} + |u|_k).$$

A similar proof for Stokes eigenvalue problem is presented by Gedicke and Khan [17]. Necessary modifications for the eigenvalue problem in elasticity with weakly imposed symmetry are done.

Proof. First, let us define

$$v = S_h(\kappa u) - u_h - (S_h(\kappa u) - u_h, u)u.$$

Applying a triangle inequality, we have

$$\begin{aligned} \|S_h(\kappa u) - u_h\|_0 &\leq \|v\|_0 + \|(S_h(\kappa u) - u_h, u)u\|_0 = \|v\|_0 + |(S_h(\kappa u) - u_h, u)| \\ &\leq \|v\|_0 + |(S_h(\kappa u) - u, u)| + |(u - u_h, u)|. \end{aligned} \quad (48)$$

By [17, Lemma 3.6], the following holds:

$$\|v\| \leq C \left(|\kappa - \kappa_h| (1 + \|S_h(\kappa u) - u_h\|_0) + \kappa h \|S_h(\kappa u) - u_h\|_0 + \|P_h u - S_h(\kappa u)\|_0 \right).$$

With Lemma 4.11, we obtain

$$\begin{aligned} \|v\| \leq C & \left(|\kappa - \kappa_h| (1 + \|S_h(\kappa u) - u_h\|_0) + \kappa h \|S_h(\kappa u) - u_h\|_0 \right. \\ & \left. + h^{k+2} (|\sigma|_{k+1} + |\gamma|_{k+1}) \right). \end{aligned}$$

We assume $\|u\|_0 = \|u_h\|_0 = 1$, then the following estimate for the third term in (48) holds:

$$|(u - u_h, u)| = 1 - (u_h, u) = \frac{1}{2} \|u - u_h\|_0^2 \leq Ch^{2k} |u|_k. \quad (49)$$

To estimate the second term, we use (36) and (38). To simplify the notation, we define $u_{\kappa,h} = S_h(\kappa u)$, $\sigma_{\kappa,h} = R_h(\kappa u)$, and $\gamma_{\kappa,h} = T_h(\kappa u)$. Choose test functions $\tau = \sigma - \sigma_{\kappa,h} \in \Sigma_g$, $v = u - u_{\kappa,h} \in U$, and $\eta = \gamma - \gamma_{\kappa,h} \in X$, such that

$$\begin{aligned} (\mathbb{C}^{-1}\sigma, \sigma - \sigma_{\kappa,h}) + (\operatorname{div}(\sigma - \sigma_{\kappa,h}), u) + (\sigma - \sigma_{\kappa,h}, \gamma) &= 0, \\ (\operatorname{div} \sigma, u - u_{\kappa,h}) &= -\kappa(u, u - u_{\kappa,h}), \\ (\sigma, \gamma - \gamma_{\kappa,h}) &= 0. \end{aligned}$$

By (36), (38), and (46), we further have

$$\begin{aligned} (\mathbb{C}^{-1}(\sigma - \sigma_{\kappa,h}), \sigma_{\kappa,h}) + (\operatorname{div} \sigma_{\kappa,h}, u - u_{\kappa,h}) + (\sigma_{\kappa,h}, \gamma - \gamma_{\kappa,h}) &= 0, \\ (\operatorname{div}(\sigma - \sigma_{\kappa,h}), u_{\kappa,h}) &= 0, \\ (\sigma - \sigma_{\kappa,h}, \gamma_{\kappa,h}) &= 0. \end{aligned}$$

Then, we obtain

$$\begin{aligned} -\kappa(u, u - u_{\kappa,h}) &= (\mathbb{C}^{-1}\sigma, \sigma - \sigma_{\kappa,h}) + (\operatorname{div}(\sigma - \sigma_{\kappa,h}), u) + (\sigma - \sigma_{\kappa,h}, \gamma) \\ &\quad + (\operatorname{div} \sigma, u - u_{\kappa,h}) + (\sigma, \gamma - \gamma_{\kappa,h}) \\ &\quad - (\mathbb{C}^{-1}(\sigma - \sigma_{\kappa,h}), \sigma_{\kappa,h}) - (\operatorname{div} \sigma_{\kappa,h}, u - u_{\kappa,h}) - (\sigma_{\kappa,h}, \gamma - \gamma_{\kappa,h}) \\ &\quad - (\operatorname{div}(\sigma - \sigma_{\kappa,h}), u_{\kappa,h}) - (\sigma - \sigma_{\kappa,h}, \gamma_{\kappa,h}). \end{aligned}$$

With (6), we get

$$\begin{aligned} -\kappa(u, u - u_{\kappa,h}) &= (\mathbb{C}^{-1}(\sigma - \sigma_{\kappa,h}), \sigma - \sigma_{\kappa,h}) + 2(\operatorname{div}(\sigma - \sigma_{\kappa,h}), u - u_{\kappa,h}) \\ &\quad + 2(\gamma - \gamma_{\kappa,h}, \sigma - \sigma_{\kappa,h}). \end{aligned}$$

Applying Cauchy-Schwarz inequality and Lemma 4.8 yields

$$\begin{aligned} -\kappa(u, u - u_{\kappa,h}) &\leq \|\sigma - \sigma_{\kappa,h}\|_{\mathbb{C}^{-1/2}}^2 + 2\|\operatorname{div}(\sigma - \sigma_{\kappa,h})\|_0 \|u - u_{\kappa,h}\|_0 \\ &\quad + 2\|\gamma - \gamma_{\kappa,h}\|_0 \|\sigma - \sigma_{\kappa,h}\|_0 \\ &\leq Ch^{2(k+1)} (|\sigma|_{k+1} + |\gamma|_{k+1}). \end{aligned}$$

Then, we obtain

$$\begin{aligned} \|S_h(\kappa u) - u_h\|_0 &\leq C \left(|\kappa - \kappa_h| (1 + \|S_h(\kappa u) - u_h\|_0) + \kappa h \|S_h(\kappa u) - u_h\|_0 \right. \\ &\quad \left. + h^{k+2} (|\sigma|_{k+1} + |\gamma|_{k+1}) + h^{2k} |u|_k \right). \end{aligned}$$

For h sufficiently small and Theorem 4.10, the term $\|S_h(\kappa u) - u_h\|_0$ on the right-hand side is absorbed and finally, we conclude for $k \geq 2$

$$\|S_h(\kappa u) - u_h\|_0 \leq Ch^{k+2} (|\sigma|_{k+1} + |\gamma|_{k+1} + |u|_k).$$

□

Remark 4.15. *In the lowest-order Falk finite element approximation ($k = 1$), by (49), we only have*

$$\|S_h(\kappa u) - u_h\|_0 \leq Ch^2(|\sigma|_2 + |\gamma|_2 + |u|_1).$$

5 Postprocessing

In this section, a postprocessing in the spirit of Stenberg [24] is presented. The particular postprocessing that we apply was first introduced by Gopalakrishnan and Guzmán [18, page 371]. On rectangular elements in elasticity with reduced symmetry, a postprocessing is discussed by Juntunen and Lee [19]. For the Stokes eigenvalue problem, a postprocessing is provided by Gedicke and Khan [17], and for the Stokes source problem by Carstensen, Gedicke, and Park [13].

First, we will introduce corresponding polynomial spaces and associated projections onto these polynomial spaces for our postprocessing. The geometry of Hilbert spaces allows a useful orthogonal decomposition.

Then, we state the local postprocessing method and discuss well-posedness of this problem. The postprocessed Falk finite element eigenfunction approximation can be found in Theorem 5.1. In addition, a new discrete eigenvalue is presented via the value of the Rayleigh quotient using the postprocessed eigenfunction. This leads to an improvement in the eigenvalue approximation error.

Recall that the space U_h consists of vector-valued square-integrable piecewise polynomials on each element with polynomial degree less than or equal to k , and we further define

$$U_h^* := \{v_h \in L^2(\Omega; \mathbb{R}^2) : v_h|_T \in \mathcal{P}_{k+1}(\mathcal{T}_h; \mathbb{R}^2)\}.$$

The space of members in U_h^* that are orthogonal to U_h is given by

$$\tilde{U}_h := \{v_h \in U_h^* : (v_h, w_h) = 0 \text{ for all } w_h \in U_h\}.$$

Hence, we have

$$U_h \perp \tilde{U}_h. \quad (50)$$

The L^2 -projection P_h of some $u \in U$ onto the discrete space U_h is the unique solution to the variational problem

$$\int_{\Omega} u \cdot w_h \, dx = \int_{\Omega} P_h u \cdot w_h \, dx \quad \text{for all } w_h \in U_h.$$

Similarly, we denote by $\tilde{P}_h$ the L^2 -projection onto $\tilde{U}_h$ and by P_h^* the L^2 -projection onto U_h^* . In general, the projections are mappings

$$\begin{aligned} P_h : U &\rightarrow U_h, & u &\mapsto P_h u, \\ \tilde{P}_h : U &\rightarrow \tilde{U}_h, & u &\mapsto \tilde{P}_h u, \\ P_h^* : U &\rightarrow U_h^*, & u &\mapsto P_h^* u. \end{aligned}$$

As already stated in [19], the identity

$$P_h^* = P_h + \tilde{P}_h \quad (51)$$

holds. The validity of this statement can be seen as follows. For any $v_h \in U_h^*$, there exist $w_h \in U_h$ and $\tilde{v}_h \in \tilde{U}_h$ such that $v_h = w_h + \tilde{v}_h$ and we have

$$\begin{aligned} (P_h^* u, v_h) &= (u, v_h) = (u, w_h + \tilde{v}_h) \\ &= (u, w_h) + (u, \tilde{v}_h) \\ &= (P_h u, w_h) + (P_h u, \tilde{v}_h) + (\tilde{P}_h u, w_h) + (\tilde{P}_h u, \tilde{v}_h) \\ &= (P_h u, v_h) + (\tilde{P}_h u, v_h) = ((P_h + \tilde{P}_h)u, v_h). \end{aligned}$$

Here we used that, by (50), $(P_h u, \tilde{v}_h) = (\tilde{P}_h u, w_h) = 0$. This is a result that follows from the geometry of Hilbert spaces. If V is a subspace in a given Hilbert space H , then it is well-known that for any $h \in H$, there exist $v \in V$ and $\tilde{v} \in V^\perp$ (the orthogonal complement of V in H) such that

$$h = v + \tilde{v}.$$

This can for example be found in [8].

5.1 Local Postprocessing

Suppose $\kappa_h \in \mathbb{R}$ and $u_h \in U_h$ solve, for some $(\sigma_h, \gamma_h) \in \Sigma_{g,h} \times X_h$, the eigenvalue problem (37). Construct an approximation $u_h^* \in U_h^*$ to u separately on each element $T \in \mathcal{T}_h$ such that

$$\begin{aligned} P_h u_h^*|_T &= u_h|_T, \\ (\nabla u_h^*, \nabla v_h)_T &= (\mathbb{C}^{-1} \sigma_h + \gamma_h, \nabla v_h)_T \quad \text{for all } v_h \in \tilde{U}_h. \end{aligned} \tag{52}$$

It is clear that the following holds:

$$(u_h^*, w_h)_T = (u_h, w_h)_T \quad \text{for all } w_h \in U_h.$$

Compatibility and well-posedness

Let us first discuss the compatibility of the problem in terms of its data. If the right-hand sides of the two equations (52) are equal to zero, then the first equation implies $u_h^* \in \tilde{U}_h$. We know $\tilde{U}_h \perp \mathcal{P}_0(T; \mathbb{R}^2)$. Therefore u_h^* would be zero and no compatibility condition is needed.

Problem (52) can be formulated as a mixed problem. For this purpose, let us introduce the Lagrange multiplier g_h and seek solutions $(u_h^*, g_h) \in U_h^* \times U_h$ locally on each element $T \in \mathcal{T}_h$ such that

$$\begin{aligned} (\nabla u_h^*, \nabla v_h)_T + (v_h, g_h)_T &= (\mathbb{C}^{-1} \sigma_h + \gamma_h, \nabla v_h)_T \quad \text{for all } v_h \in U_h^*, \\ (u_h^*, w_h)_T &= (u_h, w_h)_T \quad \text{for all } w_h \in U_h. \end{aligned} \tag{53}$$

For mixed variational problems to be well-defined, the LBB-condition must be fulfilled. Since $U_h \subset U_h^*$, we have

$$\inf_{\substack{w_h \in U_h \\ w_h \neq 0}} \sup_{\substack{v_h \in U_h^* \\ v_h \neq 0}} \frac{(v_h, w_h)_T}{\|v_h\|_{0,T} \|w_h\|_{0,T}} \geq \inf_{\substack{w_h \in U_h \\ w_h \neq 0}} \frac{(w_h, w_h)_T}{\|w_h\|_{0,T}^2} = 1.$$

The space U_h' denotes the dual space of U_h and we define the operator $A_h : U_h^* \rightarrow U_h'$ locally by

$$(A_h v_h)(w_h)|_T := (v_h, w_h)_T \quad \text{for all } v_h \in U_h^* \text{ and } w_h \in U_h.$$

Here, the duality product is just the L^2 -product. Then the kernel of A_h is given by

$$\ker A_h = \{v_h \in U_h^* : (v_h, w_h)_T = 0 \text{ for all } w_h \in U_h\}.$$

The coercivity condition of the first bilinear form in the kernel of A_h follows immediately by Poincaré's inequality.

Let us remark, $v_h \in U_h^* \cap \ker A_h$ is a polynomial of degree $k+1$ for $k \geq 1$ and, in particular, $U_h^* \cap \ker A_h \perp \mathcal{P}_0(T; \mathbb{R}^2)$. Then

$$(\nabla v_h, \nabla v_h)_T \geq C_p^{-1} \|v_h\|_{0,T}^2$$

holds for any $v_h \in U_h^* \cap \ker A_h$ with Poincaré constant $C_p > 0$.

Theorem 5.1. *For $k \geq 2$, the unique solution to (52) satisfies*

$$\|u - u_h^*\|_0 \leq Ch^{k+2}(|u|_{k+2} + |\sigma|_{k+1} + |\gamma|_{k+1}), \quad (54)$$

provided

$$\sigma \in H^{k+1}(\Omega; \mathbb{R}^{2 \times 2}), \quad u \in H^{k+2}(\Omega; \mathbb{R}^2), \quad \gamma \in L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}) \cap H^{k+1}(\Omega; \mathbb{R}^{2 \times 2}).$$

Proof. First, let us recall

$$\|P_h u - u_h\|_0 \leq Ch^{k+2}(|\sigma|_{k+1} + |\gamma|_{k+1}). \quad (55)$$

By the postprocessing (52) and (51), we have

$$P_h^* u = u_h^*, \quad P_h^* = P_h + \tilde{P}_h. \quad (56)$$

Applying a triangle inequality, we obtain

$$\|u - u_h^*\|_0 \leq \|u - P_h^* u\|_0 + \|P_h^* - u_h^*\|_0.$$

By Bramble-Hilbert, the first term becomes

$$\|u - P_h^* u\|_0 \leq Ch^{k+2}|u|_{k+2}, \quad (57)$$

and using (56), the second term is

$$\begin{aligned} \|P_h^* u - u_h^*\|_0 &= \|P_h^*(u - u_h^*)\|_0 \\ &= \|P_h(u - u_h^*)\|_0 + \|\tilde{P}_h(u - u_h^*)\|_0. \end{aligned}$$

By (52) and (55), we have

$$\begin{aligned} \|P_h(u - u_h^*)\|_0 &= \sum_{T \in \mathcal{T}_h} \left\| (P_h u - P_h u_h^*)|_T \right\|_{0,T} \\ &= \sum_{T \in \mathcal{T}_h} \left\| (P_h u - u_h)|_T \right\|_{0,T} \\ &= \|P_h u - u_h\|_0 \\ &\leq Ch^{k+2}(|\sigma|_{k+1} + |\gamma|_{k+1}). \end{aligned}$$

We know

$$\tilde{U}_h \perp \mathcal{P}_0(T; \mathbb{R}^2) \quad \text{for all } T \in \mathcal{T}_h,$$

and in particular, the space $\tilde{U}_h$ does not contain constant functions. Therefore, we can apply Poincaré's inequality separately on each element $T \in \mathcal{T}_h$,

$$\|\tilde{P}_h(u - u_h^*)\|_{0,T} \leq \frac{h_T}{\pi} \|\nabla(\tilde{P}_h(u - u_h^*))\|_{0,T}.$$

By (52) and the first equation in (36), we have

$$(\nabla(u - u_h^*), \nabla \tilde{P}_h(u - u_h^*))_T = (\mathbb{C}^{-1}(\sigma - \sigma_h) + \gamma - \gamma_h, \nabla \tilde{P}_h(u - u_h^*))_T$$

and

$$\begin{aligned} \tilde{P}_h(u - u_h^*) &= P_h^*(u - u_h^*) - P_h(u - u_h^*) \\ &= P_h^*u - u_h^* - (P_hu - u_h) \end{aligned}$$

so that we obtain

$$\begin{aligned} \|\nabla \tilde{P}_h(u - u_h^*)\|_{0,T}^2 &= (\nabla(P_h^*u - u_h^*), \nabla \tilde{P}_h(u - u_h^*))_T - (\nabla(P_hu - u_h), \nabla \tilde{P}_h(u - u_h^*))_T \\ &= (\nabla(P_h^*u - u), \nabla \tilde{P}_h(u - u_h^*))_T - (\nabla(P_hu - u_h), \nabla \tilde{P}_h(u - u_h^*))_T \\ &\quad + (\mathbb{C}^{-1}(\sigma - \sigma_h) + \gamma - \gamma_h, \nabla \tilde{P}_h(u - u_h^*))_T. \end{aligned}$$

Cauchy-Schwarz inequality yields

$$\begin{aligned} \|\nabla \tilde{P}_h(u - u_h^*)\|_{0,T}^2 &\leq \left(\|\nabla(P_h^*u - u)\|_{0,T} + \|\nabla(P_hu - u_h)\|_{0,T} \right. \\ &\quad \left. + \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_{0,T} + \|\gamma - \gamma_h\|_{0,T} \right) \|\nabla \tilde{P}_h(u - u_h^*)\|_{0,T}, \end{aligned}$$

and therefore

$$\begin{aligned} \|\tilde{P}_h(u - u_h^*)\|_{0,T} &\leq Ch_T \|\nabla \tilde{P}_h(u - u_h^*)\|_{0,T} \\ &\leq Ch_T \|\nabla(P_h^*u - u)\|_{0,T} + h_T \|\nabla(P_hu - u_h)\|_{0,T} \\ &\quad + h_T \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_{0,T} + h_T \|\gamma - \gamma_h\|_{0,T}. \end{aligned}$$

On each element $T \in \mathcal{T}_h$, define the nodal interpolant $I_h|_T : H^{k+2}(T; \mathbb{R}^2) \rightarrow U_h^*|_T$ by $v|_T \mapsto I_hv$. By [8],

$$|v - I_hv|_{j,T} \leq Ch^{k+2-j}|v|_{k+2,T} \quad \text{for all } v \in U_h^* \quad (58)$$

and applying a triangle inequality, we obtain

$$\|P_h^*u - u\|_{1,T} \leq \|P_h^*u - I_hu\|_{1,T} + \|I_hu - u\|_{1,T}.$$

An inverse estimate in the first term yields

$$\begin{aligned} \|P_h^*u - u\|_{1,T} &\leq Ch_T^{-1}(\|P_h^*u - I_hu\|_{0,T} + \|I_hu - u\|_{0,T}) \\ &\leq Ch_T^{-1}(\|P_h^*u - u\|_{0,T} + \|u - I_hu\|_{0,T}). \end{aligned}$$

By (57) and (58), we have

$$\begin{aligned} \sum_{T \in \mathcal{T}_h} h_T \|P_h^*u - u\|_{1,T} &\leq C \sum_{T \in \mathcal{T}_h} (\|P_h^*u - u\|_{0,T} + \|u - I_hu\|_{0,T}) \\ &\leq Ch^{k+2}|u|_{k+2}. \end{aligned}$$

Another inverse estimate shows

$$\begin{aligned} \sum_{T \in \mathcal{T}_h} h_T \|\nabla(P_hu - u_h)\|_{0,T} &\leq C \sum_{T \in \mathcal{T}_h} h_T h_T^{-1} \|P_hu - u_h\|_{0,T} \\ &\leq Ch^{k+2}(|\sigma|_{k+1} + |\gamma|_{k+1}). \end{aligned}$$

Furthermore, we have

$$\begin{aligned}\sum_{T \in \mathcal{T}_h} h_T \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_{0,T} &\leq Ch^{k+2} |\sigma|_{k+1}, \\ \sum_{T \in \mathcal{T}_h} h_T \|\gamma - \gamma_h\|_{0,T} &\leq Ch^{k+2} |\gamma|_{k+1}.\end{aligned}$$

Finally, we conclude

$$\|\tilde{P}_h(u - u_h^*)\|_{0,T} \leq Ch^{k+2} (\|u\|_{k+2} + |\sigma|_{k+1} + |\gamma|_{k+1}).$$

This proves the claim. $\square$

Remark 5.2. *The convergence rate for the lowest-order Falk element, $k = 1$, is one order less and we have*

$$\begin{aligned}\|u - u_h^*\| &= \mathcal{O}(h^{k+2}) \quad \text{for } k \geq 2, \\ \|u - u_h^*\| &= \mathcal{O}(h^2) \quad \text{for } k = 1.\end{aligned}$$

The restrictive term for $k = 1$ is

$$\|P_h u - u_h\|_0 = \mathcal{O}(h^2).$$

Remark 5.3. *In the postprocessing, we have assumed that the solution is sufficiently smooth. A priori, however, it is not clear whether the solution fulfils this. Nevertheless, we apply the postprocessing later in the numerical experiments, even on domains where we cannot expect the solution to be smooth. Then, of course, the stated convergence rate in the approximation is not achieved. In this case, the here postulated convergence rate can be recovered using marked meshes in the refinement instead of uniform refinement.*

Remark 5.4. *While the material tensor $\mathbb{C}$*

$$\|\mathbb{C}\epsilon\|_0 = \|\lambda\epsilon + 2\mu \operatorname{tr} \epsilon \mathbf{I}_{2 \times 2}\|_0 \longrightarrow \infty \quad \text{as } \lambda \rightarrow \infty$$

leads to Locking, its inverse $\mathbb{C}^{-1}$

$$\|\mathbb{C}^{-1}\tau\|_0 = \left\| \frac{1}{2\mu} \left(\tau - \frac{\lambda}{2(\mu + \lambda)} \operatorname{tr} \tau \mathbf{I}_{2 \times 2} \right) \right\|_0 \longrightarrow \frac{1}{2\mu} \|\operatorname{dev} \tau\|_0 \quad \text{as } \lambda \rightarrow \infty$$

does not. Hence the constant C in Theorem 5.1 does not depend on λ .

Numerical treatment

Let us briefly discuss how this method is implemented numerically. For this purpose, let $\{\varphi_j : 0 \leq j \leq k\}$ be a basis of U_h and $\{\varphi_j^* : 0 \leq j \leq k+1\}$ be a basis of U_h^* . Furthermore, introduce on each element $T \in \mathcal{T}_h$ the matrices $A_{h,T} \in \mathbb{R}^{(k+1) \times (k+1)}$ and $B_{h,T} \in \mathbb{R}^{(k+1) \times k}$ with entries

$$\begin{aligned}(A_{h,T})_{k,j} &= \int_T \nabla \varphi_{k,T}^* : \nabla \varphi_{j,T}^* \, dx, \\ (B_{h,T})_{k,j} &= \int_T \varphi_{k,T}^* \cdot \varphi_{j,T} \, dx.\end{aligned}$$

The saddle point problem (53) can numerically be solved by solving the following matrix-vector-system on each element $T \in \mathcal{T}_h$:

$$\begin{pmatrix} A_{h,T} & B_{h,T} \\ B_{h,T}^\top & 0 \end{pmatrix} \begin{pmatrix} \bar{u}_h^* \\ \bar{g}_h \end{pmatrix} = \begin{pmatrix} ((\mathbb{C}^{-1}\sigma_h + \gamma_h, \nabla \varphi_{j,T}^*)_T)_{0 \leq j \leq k+1} \\ ((u_h, \varphi_{j,T})_T)_{0 \leq j \leq k} \end{pmatrix}.$$

Here $\bar{u}_h^* \in \mathbb{R}^{k+1}$ and $\bar{g}_h \in \mathbb{R}^k$ are coefficient vectors. The number "0" at the bottom right of the matrix represents a $(k \times k)$ -matrix with zero entries. The solution u_h^* is then given by

$$u_h^* = \sum_{j=0}^{k+1} (\bar{u}_h^*)_j \varphi_j^*.$$

5.2 The postprocessed eigenvalue κ_h^*

By the second equation of (36), we have

$$(\operatorname{div} \sigma, v) = -\kappa(u, v)$$

for any $v \in U$. Using this idea, the postprocessed eigenvalue is introduced in the following definition.

Definition 5.5. *The postprocessed eigenvalue is defined as the value of the Rayleigh quotient*

$$\kappa_h^* := -\frac{(\operatorname{div} \sigma_h, u_h^*)}{(u_h^*, u_h^*)}, \quad (59)$$

provided $u_h^* \neq 0$.

Lemma 5.6. *The following identity holds:*

$$\begin{aligned} \kappa - \kappa_h^* = & \| \mathbb{C}^{-1}(\sigma - \sigma_h) + (\gamma - \gamma_h) \|_{\mathbb{C}^{1/2}}^2 - 2\mu \|\gamma - \gamma_h\|_0^2 + \kappa_h^* \|u - u_h^*\|_0^2 \\ & + 2(\operatorname{div}(\sigma - \sigma_h), u - u_h^*) - 2((\kappa_h^* - \kappa)u, u - u_h^*). \end{aligned} \quad (60)$$

Proof. Using equations (36) and (37), we have

$$\begin{aligned} \| \mathbb{C}^{-1/2}(\sigma - \sigma_h) \|_0^2 = & (\mathbb{C}^{-1}\sigma, \sigma) + (\mathbb{C}^{-1}\sigma_h, \sigma_h) - 2(\mathbb{C}^{-1}\sigma, \sigma_h) \\ = & -(\operatorname{div} \sigma, u) - (\gamma, \sigma) - (\operatorname{div} \sigma_h, u_h) - (\gamma_h, \sigma_h) \\ & + 2(\operatorname{div} \sigma_h, u) + (\gamma, \sigma_h). \end{aligned}$$

Further, let us assume $(u, u) = 1$. Again (36) and (37) yield

$$\| \mathbb{C}^{-1/2}(\sigma - \sigma_h) \|_0^2 = \kappa - (\operatorname{div} \sigma_h, u_h) + 2(\operatorname{div} \sigma_h, u) - 2(\sigma - \sigma_h, \gamma - \gamma_h).$$

Since $\operatorname{div} \sigma_h \in U_h$, the following

$$(\operatorname{div} \sigma_h, u_h) = (\operatorname{div} \sigma_h, u_h^*)$$

holds by (53), and this, together with the definition of the postprocessed eigenvalue, i.e.

$$\kappa_h^*(u_h^*, u_h^*) = -(\operatorname{div} \sigma_h, u_h^*),$$

shows

$$\|\mathbb{C}^{-1/2}(\sigma - \sigma_h)\|_0^2 = \kappa + \kappa_h^*(u_h^*, u_h^*) + 2(\operatorname{div} \sigma_h, u) - 2(\sigma - \sigma_h, \gamma - \gamma_h).$$

Adding and subtracting $2\kappa_h^*(u_h^*, u)$ yields

$$\begin{aligned} \kappa - \kappa_h^* &= \|\mathbb{C}^{-1/2}(\sigma - \sigma_h)\|_0^2 - \kappa_h^* - \kappa_h^*(u_h^*, u_h^*) + 2\kappa_h^*(u, u_h^*) \\ &\quad - 2(\operatorname{div} \sigma_h + \kappa_h^* u_h^*, u) + 2(\sigma - \sigma_h, \gamma - \gamma_h). \end{aligned}$$

With

$$\|u - u_h^*\|_0^2 = (u, u) - 2(u, u_h^*) + (u_h^*, u_h^*),$$

we have

$$\begin{aligned} \kappa - \kappa_h^* &= \|\mathbb{C}^{-1/2}(\sigma - \sigma_h)\|_0^2 + 2(\sigma - \sigma_h, \gamma - \gamma_h) - \kappa_h^* \|u - u_h^*\|_0^2 \\ &\quad - 2(\operatorname{div} \sigma_h + \kappa_h^* u_h^*, u), \end{aligned}$$

and with

$$(\operatorname{div} \sigma_h + \kappa_h^* u_h^*, u_h^*) = 0,$$

we obtain

$$\begin{aligned} \kappa - \kappa_h^* &= \|\mathbb{C}^{-1/2}(\sigma - \sigma_h)\|_0^2 + 2(\sigma - \sigma_h, \gamma - \gamma_h) - \kappa_h^* \|u - u_h^*\|_0^2 \\ &\quad - 2(\operatorname{div} \sigma_h + \kappa_h^* u_h^*, u - u_h^*). \end{aligned}$$

By

$$\begin{aligned} 2(\gamma - \gamma_h, \sigma - \sigma_h) &= 2(\mathbb{C}^{1/2}(\gamma - \gamma_h), \mathbb{C}^{-1/2}(\sigma - \sigma_h)) \\ &= \|\mathbb{C}^{-1}(\sigma - \sigma_h) + (\gamma - \gamma_h)\|_{\mathbb{C}^{1/2}}^2 - \|\mathbb{C}^{-1/2}(\sigma - \sigma_h)\|_0^2 \\ &\quad - \|\mathbb{C}^{1/2}(\gamma - \gamma_h)\|_0^2, \end{aligned}$$

we conclude

$$\begin{aligned} \kappa - \kappa_h^* &= \|\mathbb{C}^{-1}(\sigma - \sigma_h) + (\gamma - \gamma_h)\|_{\mathbb{C}^{1/2}}^2 - 2\mu \|\gamma - \gamma_h\|_0^2 - \kappa_h^* \|u - u_h^*\|_0^2 \\ &\quad - 2(\operatorname{div} \sigma_h + \kappa_h^* u_h^*, u - u_h^*). \end{aligned}$$

Adding and subtracting $2(\operatorname{div} \sigma, u - u_h^*)$ in the last term together with

$$(\operatorname{div} \sigma, u - u_h^*) = -\kappa(u, u - u_h^*),$$

which holds by (36) (since $u - u_h^* \in U$), yield

$$\begin{aligned} \kappa - \kappa_h^* &= \|\mathbb{C}^{-1}(\sigma - \sigma_h) + (\gamma - \gamma_h)\|_{\mathbb{C}^{1/2}}^2 - 2\mu \|\gamma - \gamma_h\|_0^2 + 2(\operatorname{div}(\sigma - \sigma_h), u - u_h^*) \\ &\quad - 2(\kappa_h^* u_h^* - \kappa u, u - u_h^*) - \kappa_h^* \|u - u_h^*\|_0^2 \\ &= \|\mathbb{C}^{-1}(\sigma - \sigma_h) + (\gamma - \gamma_h)\|_{\mathbb{C}^{1/2}}^2 - 2\mu \|\gamma - \gamma_h\|_0^2 + 2(\operatorname{div}(\sigma - \sigma_h), u - u_h^*) \\ &\quad + 2\kappa_h^*(u - u_h^*, u - u_h^*) - 2((\kappa_h^* - \kappa)u, u - u_h^*) - \kappa_h^* \|u - u_h^*\|_0^2 \\ &= \|\mathbb{C}^{-1}(\sigma - \sigma_h) + (\gamma - \gamma_h)\|_{\mathbb{C}^{1/2}}^2 - 2\mu \|\gamma - \gamma_h\|_0^2 + \kappa_h^* \|u - u_h^*\|_0^2 \\ &\quad + 2(\operatorname{div}(\sigma - \sigma_h), u - u_h^*) - 2((\kappa_h^* - \kappa)u, u - u_h^*). \end{aligned}$$

□

Theorem 5.7. For $k \geq 2$, assume $\kappa \in \mathbb{R}$ and $u \in H^{k+2}(\Omega; \mathbb{R}^2)$ solve, for some

$$\begin{aligned}\sigma &\in \Sigma_g \cap H^{k+1}(\Omega; \mathbb{R}^{2 \times 2}), \\ \gamma &\in X \cap H^{k+1}(\Omega; \mathbb{R}^{2 \times 2}),\end{aligned}$$

the eigenvalue problem (36). Further, assume $\kappa_h \in \mathbb{R}$ and $u \in U_h$ solve, for some $(\sigma_h, \gamma_h) \in \Sigma_{g,h} \times X_h$, the discrete eigenvalue problem (37). Moreover, let u_h^* with $u_h^* = u_h^*|_T$ for all $T \in \mathcal{T}_h$ be the solution of (52). Then the following estimate

$$|\kappa - \kappa_h^*| \leq Ch^{2k+2}(|u|_{k+2} + |\sigma|_{k+1} + |\gamma|_{k+1}) \quad (61)$$

holds for sufficiently small h .

Proof. Using Lemma 5.6 and applying Cauchy-Schwarz inequality, we obtain

$$\begin{aligned}|\kappa - \kappa_h^*| &\leq C \left(\|\sigma - \sigma_h\|_0^2 + \|\gamma - \gamma_h\|_0^2 + \|u - u_h^*\|_0^2 + \|\operatorname{div}(\sigma - \sigma_h)\|_0 \|u - u_h^*\|_0 \right. \\ &\quad \left. + 2|\kappa - \kappa_h^*| \|u - u_h^*\|_0 \right).\end{aligned}$$

Then Theorem 5.1 and (4.8) complete the proof. $\square$

Remark 5.8. In the lowest-order Falk finite element approximation, $k = 1$, we obtain

$$|\kappa - \kappa_h^*| = \mathcal{O}(h^3).$$

Here, the postprocessed eigenfunction approximation is of the same order as the stress approximation and the skew-symmetric part approximation. Therefore, we only have

$$\|\operatorname{div}(\sigma - \sigma_h)\|_0 \|u - u_h^*\|_0 = \mathcal{O}(h^3).$$

6 A posteriori error analysis

This section proposes an error estimator in the spirit of [11], and its reliability and efficiency are shown. In constructing this estimator, the primal mixed eigenvalue problem in elasticity with weakly imposed symmetry is used. For this purpose, let us briefly discuss well-posedness of the problem.

Finally, we will show that this is an error estimator for the postprocessed Falk finite element eigenvalue approximation.

Throughout this section, the notation $a \lesssim b$ abbreviates $a \leq Cb$ for some constant C , which does not depend on the mesh sizes and the Lamé parameter λ . Further, by $a \approx b$, we mean $a \lesssim b$ and $a \gtrsim b$.

Primal mixed problem in elasticity with weakly imposed symmetry

The primal mixed formulation of the eigenvalue problem in elasticity with weakly imposed symmetry is: find $(\sigma, u, \gamma) \in L^2(\Omega; \mathbb{R}^{2 \times 2}) \times H_0^1(\Omega; \mathbb{R}^2) \times L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2})$ such that

$$\begin{aligned} (\mathbb{C}^{-1}\sigma, \tau) - (\tau, \nabla u) + (\gamma, \tau) &= 0 & \text{for all } \tau \in L^2(\Omega; \mathbb{R}^{2 \times 2}), \\ (\sigma, \nabla v) &= \kappa(u, v) & \text{for all } v \in H_0^1(\Omega; \mathbb{R}^2), \\ (\sigma, \eta) &= 0 & \text{for all } \eta \in L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}). \end{aligned} \quad (62)$$

To have the problem well-posed, we need the inf-sup condition

$$\inf_{\substack{v \in H_0^1(\Omega; \mathbb{R}^2), v \neq 0 \\ \eta \in L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}), \eta \neq 0}} \sup_{\substack{\tau \in L^2(\Omega; \mathbb{R}^{2 \times 2}) \\ \tau \neq 0}} \frac{-(\tau, \nabla v) + (\tau, \eta)}{\|\tau\|_0 (\|\nabla v\|_0 + \|\eta\|_0)} \geq C > 0 \quad (63)$$

fulfilled. This does not simply follow by the inf-sup condition of the primal mixed problem in elasticity with symmetric stress tensor σ . For symmetric stress, the inf-sup condition follows immediately, i.e. for given $v \in H_0^1(\Omega)$, we have $\tau = \epsilon(v) \in L^2(\Omega; \mathbb{R}^{2 \times 2})$ and

$$\sup_{\substack{\tau \in L^2(\Omega; \mathbb{R}^{2 \times 2}) \\ \tau \neq 0}} \frac{(\tau, \epsilon(v))}{\|\tau\|_0 \|\epsilon(v)\|_0} \geq \frac{(\epsilon(v), \epsilon(v))}{\|\epsilon(v)\|_0^2} = 1.$$

To show the inf-sup condition (63), we use Helmholtz decomposition

$$L^2(\Omega; \mathbb{R}^2) = \nabla H_0^1(\Omega) \oplus \text{Curl } H^1(\Omega),$$

which is an orthogonal decomposition. For any $f \in L^2(\Omega; \mathbb{R}^2)$ exists a unique potential field $\varphi \in H_0^1(\Omega)$ and a unique stream function $\psi \in H^1(\Omega)$ such that

$$f = \nabla \varphi + \text{Curl } \psi.$$

Then for given $v \in H_0^1(\Omega; \mathbb{R}^2)$, we construct some $\psi \in H^1(\Omega; \mathbb{R}^2)$ as a solution to an auxiliary Stokes problem such that

$$\begin{aligned} -\nabla v + \text{Curl } \psi &\in L^2(\Omega; \mathbb{R}^{2 \times 2}), \\ (\nabla v, \text{Curl } \psi) &= 0. \end{aligned} \quad (64)$$

Find $(\psi, p) \in H_0^1(\Omega; \mathbb{R}^2) \times L^2(\Omega)/\mathbb{R}$ satisfying

$$\begin{aligned} 2\mu(\epsilon(\psi), \epsilon(\varphi)) - (p, \operatorname{div} \varphi) &= 0 & \text{for all } \varphi \in H_0^1(\Omega; \mathbb{R}^2), \\ -(w, \operatorname{div} \psi) &= -(\operatorname{curl} v + q, w) & \text{for all } w \in L^2(\Omega)/\mathbb{R}. \end{aligned} \quad (65)$$

For $q \in L^2(\Omega)$, let $\eta \in L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2})$ be such that

$$\eta = S^2(q).$$

By the second equation in (65), we have

$$(q, q) = -(\operatorname{curl} v, q) + (q, \operatorname{div} \psi).$$

Further, (28) yields

$$(q, q) = -(\operatorname{curl} v, q) + (S^2(q), \operatorname{Curl} \psi)$$

and

$$\begin{aligned} \frac{1}{2}(\eta, \eta) &= \frac{1}{2}(S^2(q), S^2(q)) = (q, q) = -(\operatorname{curl} v, q) + (\operatorname{Curl} \psi, \eta) \\ &= -(\nabla v, \eta) + (\operatorname{Curl} \psi, \eta). \end{aligned} \quad (66)$$

Here, the last equality is valid by

$$\int_{\Omega} \nabla v : S^2(q) \, dx = \int_{\Omega} \begin{pmatrix} \partial_1 v_1 & \partial_2 v_1 \\ \partial_1 v_2 & \partial_2 v_2 \end{pmatrix} : \begin{pmatrix} 0 & -q \\ q & 0 \end{pmatrix} \, dx = \int_{\Omega} q \operatorname{curl} v \, dx.$$

We choose $\tau = -\nabla v + \operatorname{Curl} \psi$ and by (64), we obtain

$$\begin{aligned} \sup_{\substack{\tau \in L^2(\Omega; \mathbb{R}^{2 \times 2}) \\ \tau \neq 0}} \frac{-(\tau, \nabla v) + (\tau, \eta)}{\|\tau\|_0(\|\nabla v\|_0 + \|\eta\|_0)} &\geq \frac{(\nabla v - \operatorname{Curl} \psi, \nabla v) + (-\nabla v + \operatorname{Curl} \psi, \eta)}{\|-\nabla v + \operatorname{Curl} \psi\|_0(\|\nabla v\|_0 + \|\eta\|_0)} \\ &= \frac{\|\nabla v\|_0^2 + \frac{1}{2}\|\eta\|_0^2}{\|-\nabla v + \operatorname{Curl} \psi\|_0(\|\nabla v\|_0 + \|\eta\|_0)} \\ &\gtrsim \frac{\|\nabla v\|_0^2 + \|\eta\|_0^2}{\|\nabla v\|_0(\|\nabla v\|_0 + \|\eta\|_0)} \geq C > 0. \end{aligned}$$

In this formulation, the stress is only in $L^2(\Omega; \mathbb{R}^{2 \times 2})$ with generally no square-integrable divergence. For the primal problem in elasticity with symmetric stress tensor σ , the coercivity condition is discussed in [6]. The coercivity condition for the primal problem in elasticity with reduced symmetry is similar. First, we define the following space in which the divergence is understood in the distributional sense:

$$\begin{aligned} Z &= \{\tau \in L^2(\Omega; \mathbb{R}^{2 \times 2}) : \exists h \in (C_c^\infty(\Omega; \mathbb{R}^2))' \text{ s.t. } h = \operatorname{div} \tau \text{ and} \\ &\quad \langle h, \varphi \rangle = -\langle \tau, \nabla \varphi \rangle \forall \varphi \in C_c^\infty(\Omega; \mathbb{R}^2)\}. \end{aligned}$$

Here $\langle \bullet, \bullet \rangle$ denotes the duality product. Let $\tilde{B} : L^2(\Omega; \mathbb{R}^{2 \times 2}) \rightarrow \mathbb{R}$ defined by

$$(\tilde{B}\tau)(v) = (\tau, \nabla v).$$

The kernel of $\tilde{B}$, where we understand $\operatorname{div} \tau$ in the distributional sense, is given by

$$\ker \tilde{B} = \{\tau \in Z : \operatorname{div} \tau = 0\}.$$

For any $\tau \in \ker \tilde{B} \cap L_0^2(\Omega; \mathbb{R}^{2 \times 2})$, the coercivity condition follows by (16) independent of the Lamé parameter λ , and we have

$$(\mathbb{C}^{-1}\tau, \tau) = \|\tau\|_{\mathbb{C}^{-\frac{1}{2}}}^2 \geq C(\mu)\|\tau\|_0^2.$$

An error estimator for the postprocessed eigenvalue approximation

In Gedicke and Khan [17], an error estimator for the improved eigenvalue approximation for the Stokes eigenvalue problem has been presented and for the Stokes source problem, an error estimator has been presented by Carstensen, Gedicke, and Park [13]. Similarly, we will propose an error estimator for the postprocessed eigenvalue approximation in elasticity with reduced symmetry.

Let $\kappa \in \mathbb{R}$ and $u \in H_0^1(\Omega; \mathbb{R}^2)$ solve, for some

$$(\sigma, \gamma) \in L^2(\Omega; \mathbb{R}^{2 \times 2}) \times L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}),$$

the primal mixed eigenvalue problem (62). Further, let us define

$$W := L^2(\Omega; \mathbb{R}^{2 \times 2}) \times H_0^1(\Omega; \mathbb{R}^2) \times L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}).$$

Following the theory in [11], the operator $A : W \rightarrow W'$,

$$(A(\sigma, u, \gamma))(\tau, v, \eta) := (\mathbb{C}^{-1}\sigma, \tau) - (\tau, \nabla u) + (\gamma, \tau) + (\sigma, \nabla v) + (\sigma, \eta)$$

is linear, bounded, and bijective with bounded inverse. If

$$(\sigma_h, \tilde{u}_h, \gamma_h) \in L^2(\Omega; \mathbb{R}^{2 \times 2}) \times H_0^1(\Omega; \mathbb{R}^2) \times L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2})$$

is an approximation of (62) with right-hand side κu , then the following holds:

$$\begin{aligned} & \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0 + \|\nabla u - \nabla \tilde{u}_h\|_0 + \|\gamma - \gamma_h\|_0 \\ & \approx \|\text{Res}_1\|_{(L^2(\Omega; \mathbb{R}^{2 \times 2}))'} + \|\text{Res}_2\|_{(H_0^1(\Omega; \mathbb{R}^2))'} + \|\text{Res}_3\|_{(L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}))'}. \end{aligned}$$

The residuals are defined by

$$\begin{aligned} \text{Res}_1(\tau) &:= (\mathbb{C}^{-1}(\sigma_h), \tau) - (\tau, \nabla u_h) + (\gamma_h, \tau) & \text{for all } \tau \in L^2(\Omega; \mathbb{R}^{2 \times 2}), \\ \text{Res}_2(v) &:= -(\sigma_h, \nabla v) + \kappa(u, v) & \text{for all } v \in H_0^1(\Omega; \mathbb{R}^2), \\ \text{Res}_3(\eta) &:= (\sigma_h, \eta) & \text{for all } \eta \in L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}), \end{aligned}$$

and corresponding norms are

$$\begin{aligned} \|\text{Res}_1\|_{(L^2(\Omega; \mathbb{R}^{2 \times 2}))'} &= \sup_{\substack{\tau \in L^2(\Omega; \mathbb{R}^{2 \times 2}) \\ \tau \neq 0}} \frac{\text{Res}_1(\tau)}{\|\tau\|_0}, \\ \|\text{Res}_2\|_{(H_0^1(\Omega; \mathbb{R}^2))'} &= \sup_{\substack{v \in H_0^1(\Omega; \mathbb{R}^2) \\ v \neq 0}} \frac{\text{Res}_2(v)}{\|\nabla v\|_0}, \\ \|\text{Res}_3\|_{(L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}))'} &= \sup_{\substack{\eta \in L_{\text{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}) \\ \eta \neq 0}} \frac{\text{Res}_3(\eta)}{\|\eta\|_0}. \end{aligned}$$

We will frequently refer to terms that, for sufficiently small h , converge faster than others as higher-order terms and denote them by h.o.t..

The following Lemma provides a lower bound for the sum of the three residuals in their corresponding operator norms.

Lemma 6.1. *The following estimate holds:*

$$\begin{aligned} & \|\mathbb{C}^{-1}(\sigma - \sigma_h)\| + \|\nabla(u - \tilde{u}_h)\| + \|\gamma - \gamma_h\| \\ & \lesssim \|\mathbb{C}^{-1}\sigma_h + \gamma_h - \nabla u_h\| + \left(\sum_{K \in \mathcal{T}_h} h_T^2 \|\kappa u + \operatorname{div} \sigma_h\|^2 \right)^{1/2} + \|\operatorname{skw} \sigma_h\| + \text{h.o.t.}, \end{aligned}$$

with higher order terms

$$\text{h.o.t.} := \kappa_h \|u - u_h^*\| + |\kappa - \kappa_h|.$$

Proof. The first residual can be estimated using the Cauchy-Schwarz inequality, and we have

$$\operatorname{Res}_1(\tau) = \int_{\Omega} (\mathbb{C}^{-1}\sigma_h + \gamma_h - \nabla u_h) : \tau \, dx \leq \|\mathbb{C}^{-1}\sigma_h + \gamma_h - \nabla u_h\|_0 \|\tau\|_0.$$

Applying Gauss theorem, we obtain for any $v \in H_0^1(\Omega; \mathbb{R}^2)$

$$\operatorname{Res}_2(v) = -(\sigma_h, \nabla v) + \kappa(u, v) = (\kappa u + \operatorname{div} \sigma_h, v).$$

The resulting boundary term vanishes since $v = 0$ on $\partial\Omega$. Let v_h be the Scott-Zhang interpolation [23] of v . By linearity, (37) and the postprocessing (52), we have

$$\begin{aligned} \operatorname{Res}_2(v) &= (\kappa u + \operatorname{div} \sigma_h, v - v_h) + (\kappa u + \operatorname{div} \sigma_h, v_h) \\ &= (\kappa u + \operatorname{div} \sigma_h, v - v_h) + (\kappa u - \kappa_h \tilde{u}_h, v_h) \\ &= (\kappa u + \operatorname{div} \sigma_h, v - v_h) + (\kappa u - \kappa_h u_h^*, v_h) \\ &= (\kappa u + \operatorname{div} \sigma_h, v - v_h) + (\kappa - \kappa_h)(u, v_h) + \kappa_h(u - u_h^*, v_h). \end{aligned}$$

The Cauchy-Schwarz inequality yields

$$\begin{aligned} \operatorname{Res}_2(v) &= \int_{\Omega} (\kappa u + \operatorname{div} \sigma_h)(v - v_h) \, dx + (\kappa - \kappa_h)(u, v_h) + \kappa_h(u - u_h^*, v_h) \\ &= \sum_{T \in \mathcal{T}_h} \int_T h_T (\kappa u + \operatorname{div} \sigma_h) h_T^{-1} (v - v_h) \, dx \\ &\quad + (\kappa - \kappa_h)(u, v_h) + \kappa_h(u - u_h^*, v_h) \\ &\lesssim \left(\sum_{T \in \mathcal{T}_h} h_T^2 \|\kappa u + \operatorname{div} \sigma_h\|_{0,T}^2 \right)^{1/2} \left(\sum_{T \in \mathcal{T}_h} h_T^{-2} \|v - v_h\|_{0,T}^2 \right)^{1/2} \\ &\quad + (\kappa_h \|u - u_h^*\|_0 + |\kappa - \kappa_h|) \|v_h\|_0. \end{aligned}$$

Stability of v_h and interpolation estimates

$$\sum_{T \in \mathcal{T}_h} h_{0,T}^{-2} \|v - v_h\|_T^2 \lesssim \sum_{T \in \mathcal{T}_h} |v|_{1,T}^2 = |v|_1^2$$

yield

$$\operatorname{Res}_2(v) \lesssim \left(\left(\sum_{T \in \mathcal{T}_h} h_T^2 \|\kappa u + \operatorname{div} \sigma_h\|_{0,T}^2 \right)^{1/2} + \text{h.o.t.} \right) |v|_1.$$

Let $\eta \in L_{\operatorname{skw}}^2(\Omega; \mathbb{R}^{2,2})$, then the last residual is estimated using the Cauchy-Schwarz inequality as follows:

$$\operatorname{Res}_3(\eta) = \int_{\Omega} \sigma_h : \eta \, dx \lesssim \|\operatorname{skw}(\sigma_h)\|_0 \|\eta\|_0.$$

This proves the claim. $\square$

Since $\tilde{u}_h \in H_0^1(\Omega; \mathbb{R}^2)$, $\tilde{u}_h$ is a conforming approximation of the continuous solution u . We construct in the following a conforming approximation $\tilde{u}_h$ using the postprocessed eigenfunction u_h^* , which is in general discontinuous. Let $\tilde{u}_h$ be defined by the vertex and edge degrees of freedom in $z \in \mathbb{R}^2$ as the arithmetic mean value of u_h^*

$$\tilde{u}_h(z) := \frac{1}{|\{T \in \mathcal{T}_h : z \in T\}|} \sum_{T \in \mathcal{T}_h : z \in T} u_h^*(z)|_T.$$

The following approximation holds by [20, Theorem 2.2]

$$\sum_{T \in \mathcal{T}_h} h_T^{-2} \|u_h^* - \tilde{u}_h\|_{0,T}^2 + \sum_{T \in \mathcal{T}_h} \|\nabla_{\mathcal{T}_h} u_h^* - \nabla \tilde{u}_h\|_{0,T}^2 \lesssim \sum_{E \in \mathcal{E}_h} h_E^{-1} \|[u_h^*]\|_{0,E}^2. \quad (67)$$

We define the error estimator η by

$$\begin{aligned} \eta^2 := & \|\mathbb{C}^{-1} \sigma_h + \gamma_h - \nabla_{\mathcal{T}_h} u_h^*\|_0^2 + \sum_{T \in \mathcal{T}_h} h_T^2 \|\kappa_h^* u_h^* + \operatorname{div} \sigma_h\|_{0,T}^2 \\ & + \sum_{E \in \mathcal{E}_h} h_E^{-1} \|[u_h^*]\|_{0,E}^2 + \|\operatorname{skw}(\sigma_h)\|_0^2. \end{aligned}$$

The error estimator η is reliable and efficient

Let us first prove the following reliability estimate.

Theorem 6.2. *Let $\kappa \in \mathbb{R}_+$ and $u \in H_0^1(\Omega; \mathbb{R}^2)$, for some*

$$(\sigma, \gamma) \in L^2(\Omega; \mathbb{R}^{2,2}) \times L_{\operatorname{skw}}^2(\Omega; \mathbb{R}^{2 \times 2}),$$

solve (62). The postprocessed eigenfunction $u_h^ \in U_h^*$ satisfies*

$$\|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0 + \|\nabla u - \nabla_{\mathcal{T}_h} u_h^*\|_0 + \|\gamma - \gamma_h\|_0 \lesssim \eta + \text{h.o.t.} \quad (68)$$

with higher-order terms

$$\text{h.o.t.} = \left(\sum_{T \in \mathcal{T}_h} h_T^2 \|\kappa u - \kappa_h^* u_h^*\|_0^2 \right)^{1/2} + \kappa_h \|u - u_h^*\|_0 + |\kappa - \kappa_h|.$$

Proof. Let us first remark, by Jensen's inequality, we have

$$\begin{aligned} & \frac{1}{3} \left(\|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0 + \|\nabla u - \nabla_{\mathcal{T}_h} u_h^*\|_0 + \|\gamma - \gamma_h\|_0 \right) \\ & \leq \left(\|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0^2 + \|\nabla u - \nabla_{\mathcal{T}_h} u_h^*\|_0^2 + \|\gamma - \gamma_h\|_0^2 \right)^{1/2}. \end{aligned}$$

In the second term of (68), by adding and subtracting $\nabla \tilde{u}_h$, we obtain

$$\begin{aligned} & \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0 + \|\nabla u - \nabla_{\mathcal{T}_h} u_h^*\|_0 + \|\gamma - \gamma_h\|_0 \\ & \leq \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0 + \|\nabla(u - \tilde{u}_h)\|_0 + \|\nabla \tilde{u}_h - \nabla_{\mathcal{T}_h} u_h^*\|_0 + \|\gamma - \gamma_h\|_0 \end{aligned}$$

and applying Lemma 6.1 yields

$$\begin{aligned} & \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0 + \|\nabla u - \nabla_{\mathcal{T}_h} u_h^*\|_0 + \|\gamma - \gamma_h\|_0 \\ & \lesssim \|\mathbb{C}^{-1} \sigma_h + \gamma_h - \nabla_{\mathcal{T}_h} u_h^*\|_0 + \|\nabla \tilde{u}_h - \nabla_{\mathcal{T}_h} u_h^*\|_0 + \|\operatorname{skw} \sigma_h\|_0 \\ & \quad + \left(\sum_{T \in \mathcal{T}_h} h_T^2 \|\kappa u + \operatorname{div} \sigma_h\|_{0,T}^2 \right)^{1/2} + \kappa_h \|u - u_h^*\|_0 + |\kappa - \kappa_h|. \end{aligned}$$

Adding and subtracting $\kappa_h^* u_h^*$ gives

$$\begin{aligned}
& \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0 + \|\nabla u - \nabla_{\mathcal{T}_h} u_h^*\|_0 + \|\gamma - \gamma_h\|_0 \\
& \lesssim \|\mathbb{C}^{-1}\sigma_h + \gamma_h - \nabla_{\mathcal{T}_h} u_h^*\|_0 + \|\nabla \tilde{u}_h - \nabla_{\mathcal{T}_h} u_h^*\|_0 + \|\text{skw } \sigma_h\|_0 \\
& \quad + \left(\sum_{T \in \mathcal{T}_h} h_T^2 \|\kappa_h^* u_h^* + \text{div } \sigma_h\|_{0,T}^2 \right)^{1/2} + \left(\sum_{T \in \mathcal{T}_h} h_T^2 \|\kappa u - \kappa_h^* u_h^*\|_{0,T}^2 \right)^{1/2} \\
& \quad + \kappa_h \|u - u_h^*\|_0 + |\kappa - \kappa_h|,
\end{aligned}$$

where the last three terms are of higher order. Applying Jensen's inequality, we obtain

$$\begin{aligned}
& \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0 + \|\nabla u - \nabla_{\mathcal{T}_h} u_h^*\|_0 + \|\gamma - \gamma_h\|_0 \\
& \lesssim \left(\|\mathbb{C}^{-1}\sigma_h + \gamma_h - \nabla_{\mathcal{T}_h} u_h^*\|_0^2 + \|\nabla \tilde{u}_h - \nabla_{\mathcal{T}_h} u_h^*\|_0^2 + \sum_{K \in \mathcal{T}_h} h_K^2 \|\kappa_h^* u_h^* + \text{div } \sigma_h\|_0^2 \right. \\
& \quad \left. + \|\text{skw } \sigma_h\|_0^2 \right)^{1/2} + \text{h.o.t.} \\
& \lesssim \eta + \text{h.o.t.},
\end{aligned}$$

which completes the proof. $\square$

We continue with the following efficiency estimate.

Theorem 6.3. *Let for $\kappa_h \in \mathbb{R}_+$ and $u_h \in U_h$ solve, for some $(\sigma_h, \gamma_h) \in \Sigma_{g,h} \times X_h$, the eigenvalue problem (37). Moreover, let $u_h^* \in U_h^*$ be the postprocessed eigenfunction and κ_h^* the postprocessed eigenvalue. For sufficiently small h , we obtain*

$$\eta \lesssim \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0 + \|\gamma - \gamma_h\|_0 + \|u - u_h^*\|_0^2 + h|\kappa - \kappa_h^*| \|u\|_0.$$

Proof. The error estimator η is given by

$$\begin{aligned}
\eta^2 = & \|\mathbb{C}^{-1}\sigma_h + \gamma_h - \nabla_{\mathcal{T}_h} u_h^*\|_0^2 + \sum_{T \in \mathcal{T}_h} h_T^2 \|\kappa_h^* u_h^* + \text{div } \sigma_h\|_{0,T}^2 \\
& + \sum_{E \in \mathcal{E}_h} h_E^{-1} \|[u_h^*]\|_{0,E}^2 + \|\text{skw}(\sigma_h)\|_0^2.
\end{aligned}$$

Adding and subtracting $\mathbb{C}^{-1}\sigma + \gamma = \nabla u$ in the first term and applying the triangle inequality, we obtain

$$\|\mathbb{C}^{-1}\sigma_h + \gamma_h - \nabla_{\mathcal{T}_h} u_h^*\|_0^2 \leq \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0^2 + \|\gamma - \gamma_h\|_0^2 + \|\nabla u - \nabla_{\mathcal{T}_h} u_h^*\|_0^2.$$

In the second term, adding and subtracting $\text{div } \sigma = -\kappa u$ and applying the triangle inequality, we have

$$\begin{aligned}
h_T^2 \|\kappa_h^* u_h^* + \text{div } \sigma_h\|_{0,T}^2 & \leq h_T^2 \|\kappa_h^* u_h^* - \kappa u\|_{0,T}^2 + h_T^2 \|\text{div}(\sigma - \sigma_h)\|_{0,T}^2 \\
& \leq h_T^2 |\kappa - \kappa_h^*|^2 \|u\|_{0,T}^2 + h_T^2 (\kappa_h^*)^2 \|u - u_h^*\|_0^2 \\
& \quad + h_T^2 \|\text{div}(\sigma - \sigma_h)\|_{0,T}^2.
\end{aligned}$$

Moreover, we obtain with $\sigma_h = \text{sym } \sigma_h + \text{skw } \sigma_h$ the following:

$$\|\text{skw } \sigma_h\|_0^2 = \|\sigma - \sigma_h\|_0^2 + \|\sigma - \text{sym } \sigma_h\|_0^2 \lesssim \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0^2.$$

Lastly, by $[u] = 0$, we conclude

$$\eta^2 \lesssim \|\mathbb{C}^{-1}(\sigma - \sigma_h)\|_0^2 + \|\gamma - \gamma_h\|_0^2 + \|u - u_h^*\|_0^2 + h_T^2 |\kappa - \kappa_h^*|^2 \|u\|_0^2.$$

□

Using the results discussed above, we complete this section by showing that η is a reliable and efficient error estimator for the postprocessed eigenvalue approximation $|\kappa - \kappa_h^*|$.

Theorem 6.4. *The following estimate*

$$|\kappa - \kappa_h^*| \lesssim \eta^2 + \text{h.o.t.}$$

holds.

Proof. By Lemma 5.6, adding and subtracting $\tilde{u}_h$ as well as $\text{div } \sigma$, we get

$$\begin{aligned} \kappa - \kappa_h^* &= \|\mathbb{C}^{-1}(\sigma - \sigma_h) + \gamma - \gamma_h\|_{\mathbb{C}^{1/2}}^2 - 2\mu\|\gamma - \gamma_h\|_0^2 - \kappa_h^*\|u - u_h^*\|_0^2 \\ &\quad - 2(\text{div } \sigma_h + \kappa_h^* u_h^*, u - \tilde{u}_h) - 2(\text{div } \sigma_h + \kappa_h^* u_h^*, \tilde{u}_h - u_h^*) \\ &= \|\mathbb{C}^{-1}(\sigma - \sigma_h) + \gamma - \gamma_h\|_{\mathbb{C}^{1/2}}^2 - 2\mu\|\gamma - \gamma_h\|_0^2 - \kappa_h^*\|u - u_h^*\|_0^2 \\ &\quad + 2(\text{div } (\sigma - \sigma_h), u - \tilde{u}_h) - 2(\kappa_h^* u_h^* - \kappa u, u - \tilde{u}_h) \\ &\quad - 2(\text{div } \sigma_h + \kappa_h^* u_h^*, \tilde{u}_h - u_h^*). \end{aligned}$$

Due to boundary conditions, we have $(u - \tilde{u}_h) \cdot n = 0$ on Γ_D and $(\sigma - \sigma_h) \cdot n = 0$ on Γ_N . Applying Gauss theorem yields

$$\begin{aligned} \kappa - \kappa_h^* &= \|\mathbb{C}^{-1}(\sigma - \sigma_h) + \gamma - \gamma_h\|_{\mathbb{C}^{1/2}}^2 - 2\mu\|\gamma - \gamma_h\|_0^2 - \kappa_h^*\|u - u_h^*\|_0^2 \\ &\quad - 2(\sigma - \sigma_h, \nabla(u - \tilde{u}_h)) + 2\kappa_h^*(u - u_h^*, u - \tilde{u}_h) \\ &\quad + 2(\kappa - \kappa_h^*)(u, u - \tilde{u}_h) - 2(\text{div } \sigma_h + \kappa_h^* u_h^*, \tilde{u}_h - u_h^*). \end{aligned}$$

and the boundary integral vanishes. Using the triangle inequality and Young's inequality, we obtain

$$\begin{aligned} |\kappa - \kappa_h^*| &\lesssim \|\mathbb{C}^{-1/2}(\sigma - \sigma_h)\|_0^2 + \|\gamma - \gamma_h\|_0^2 + \|\nabla_{\mathcal{T}_h} u_h^* - \nabla \tilde{u}_h\|_0^2 \\ &\quad + \|\nabla u - \nabla_{\mathcal{T}_h} u_h^*\|_0^2 + \sum_{T \in \mathcal{T}_h} \left(h_T^2 \|\text{div } \sigma_h + \kappa_h^* u_h^*\|_{0,T}^2 + h_T^{-2} \|\tilde{u}_h - u_h^*\|_{0,T}^2 \right) \\ &\quad + \|u_h^* - \tilde{u}_h\|_0^2 + \|u - u_h^*\|_0^2 + (\kappa - \kappa_h^*)^2. \end{aligned}$$

Finally, for h small enough, with Theorem 6.2 and (67), we conclude

$$|\kappa - \kappa_h^*| \lesssim \eta^2 + \text{h.o.t.}$$

holds.

□

Therefore our error estimator η is an upper bound for our postprocessed eigenvalue approximation error, provided the mesh size h is small enough. For a particular u_h^* , the efficiency follows by Theorem 5.1 and Theorem 6.3.

7 Numerical results

In this section, we present some numerical results on the square domain, L-shaped domain, and Cook's Membrane that support our theoretical results. In all of our experiments, we choose the Lamé parameters as follows. The value for λ is either 1 or 10 or 100, and $\mu = 1$.

The experiments are performed on uniformly refined and on adaptively refined meshes, respectively. For the adaptive refinement, we are using the standard finite element loop. Let $\mathcal{T}_0$ be an initial refinement of $\Omega \in \mathbb{R}^2$. Then

$$\text{Solve} \rightarrow \text{Estimate} \rightarrow \text{Mark} \rightarrow \text{Refine}$$

generates a sequence of meshes $\mathcal{T}_1, \mathcal{T}_2, \mathcal{T}_3 \dots$ and a sequence of corresponding discrete spaces $\Sigma_{g,h}^j$, U_h^j , and X_h^j for $j \in \mathbb{N}$. The eigenvalue approximation error is approximated on various levels j by the difference between the discrete eigenvalue and a reference value. Let κ be the reference eigenvalue, which is exact in case it is known and a high accuracy approximation in case it is unknown. Further, let κ_h be the computed eigenvalue by solving (37) and κ_h^* be the computed postprocessed eigenvalue as in Definition 5.5. To avoid confusion, let us call $|\kappa - \kappa_h|$ the eigenvalue approximation error and $|\kappa - \kappa_h^*|$ the postprocessed eigenvalue approximation error in the following.

All graphs are plotted in double logarithmic scale so that the convergence rate is twice the negative slope.

We assume that the eigenvalues are in ascending order in the spectrum, i.e.

$$\kappa_h^1 \leq \kappa_h^2 \leq \kappa_h^3 \leq \dots$$

Let $\kappa_h \in \mathbb{R}$ and $u_h \in U_h$ solve, for some $\sigma_h \in \Sigma_{g,h}$ and $\gamma_h \in X_h$, the eigenvalue problem (37). Moreover, let u_h^* be the solution of (52). Then we compute the error estimator η by

$$\begin{aligned} \eta^2 = & \|\mathbb{C}^{-1}\sigma_h + \gamma_h - \nabla_{\mathcal{T}_h} u_h^*\|_0^2 + \sum_{T \in \mathcal{T}_h} h_T^2 \|\lambda_h^* u_h^* + \text{div } \sigma_h\|_{0,T}^2 \\ & + \sum_{E \in \mathcal{E}} h_E^{-1} \|[u_h^*]\|_{0,E}^2 + \|\text{skw}(\sigma_h)\|_0^2. \end{aligned}$$

From [12, (1.10)], an error estimator μ for the eigenvalue approximation $|\kappa - \kappa_h|$ is presented with some modifications and without proof. The originally stated estimator, which we call $\tilde{\mu}$, is given by the following:

$$\begin{aligned} \tilde{\mu}^2 = & \sum_{T \in \mathcal{T}_h} h_T^2 \|\text{div}(\sigma - \sigma_h)\|_{0,T}^2 + \|\text{skw}(\sigma_h)\|_{0,T}^2 \\ & + \sum_{E \in \mathcal{E}_I \cup \mathcal{E}_D} h_E \|\mathbb{C}^{-1}\sigma_h + \gamma_h\|_{0,E}^2 + \sum_{E \in \mathcal{E}_N} h_E \|[\sigma - \sigma_h] \cdot n_E\|_{0,E}^2 \\ & + \inf_{v_h \in U_h} \sum_{T \in \mathcal{T}_h} h_T^2 \|\mathbb{C}^{-1}\sigma_h + \gamma_h - \nabla_{\mathcal{T}_h} v_h\|_{0,T}^2. \end{aligned}$$

The term

$$\sum_{T \in \mathcal{T}_h} h_T^2 \|\text{div } \sigma + \lambda_h^* u_h^*\|_0^2 = \sum_{T \in \mathcal{T}_h} h_T^2 \|\lambda_h^* u_h^* - \lambda u\|_0^2$$

is of higher-order and $\sigma \cdot n = 0$ on Γ_N . Using the eigenvalue problem (36) and a triangle inequality, we define the new estimator μ with some modification from $\tilde{\mu}$ by

$$\begin{aligned} \mu^2 = & \sum_{T \in \mathcal{T}_h} h_T^2 \|\lambda_h^* u_h^* + \operatorname{div} \sigma_h\|_{0,T}^2 + \|\operatorname{skw}(\sigma_h)\|_{0,T}^2 + h_T^2 \|\mathbb{C}^{-1} \sigma_h + \gamma_h - \nabla_{\mathcal{T}_h} u_h\|_{0,T}^2 \\ & + \sum_{E \in \mathcal{E}_I \cup \mathcal{E}_D} h_E \|\mathbb{C}^{-1} \sigma_h + \gamma_h\|_{0,E}^2 + \sum_{E \in \mathcal{E}_N} h_E \|\sigma_h \cdot n_E\|_{0,E}^2. \end{aligned}$$

We are using the MatLab implementation *fem4m* of the Group of J. Gedicke [16]. In the implementation, the eigenvalue problem is solved using the in MatLab pre-implemented eigs solver.

7.1 Square domain

As a first and numerical example, let us consider the square domain $\Omega = (0, 1)^2$. By [15], we know the exact eigenvalues for symmetric boundary conditions. For pure Neumann boundary conditions, some exact eigenvalues could be found, and are displayed in Table 1.

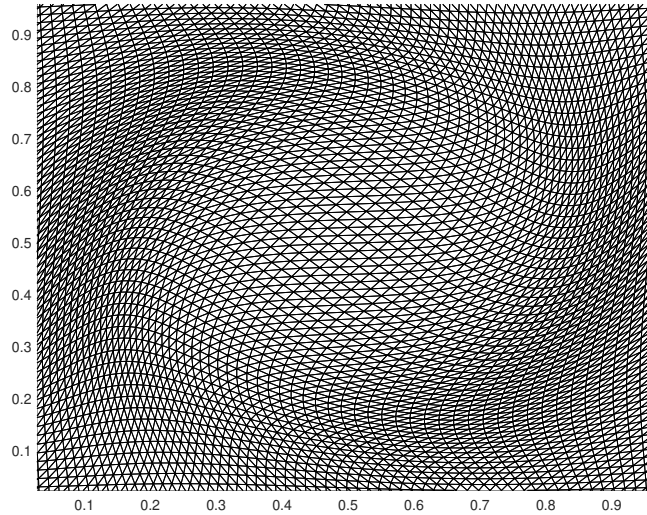


Figure 5: The third eigenfunction on an uniformly refined mesh of the square domain with Dirichlet boundary conditions.

Dirichlet Boundary conditions

The exact eigenvalues for homogeneous Dirichlet boundary conditions are unknown. We use the reference eigenvalue $\kappa = 51.294997977322$, which is a high accuracy approximation of the third eigenvalue calculated using higher-order polynomials.

In the a priori analysis, we have assumed that κ is a simple eigenvalue, although we know that multiple eigenvalues also exist in the spectrum whose eigenfunctions jointly span the associated eigenspace. Indeed, the first two eigenvalues in the spectrum are multiple ones.

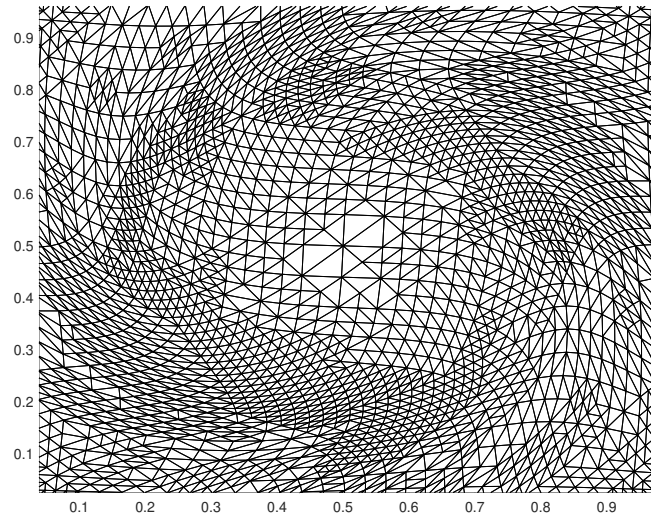


Figure 6: The third eigenfunction on an adaptively refined mesh of the square domain with Dirichlet boundary conditions.

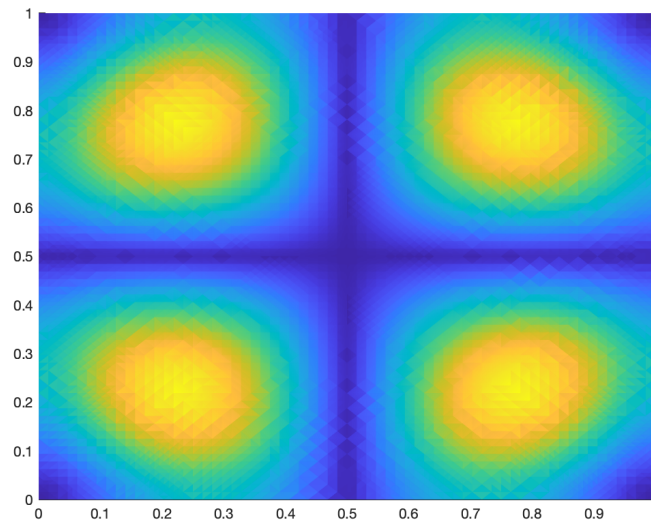


Figure 7: Von Mises stress plot to the third eigenvalue on the square domain with Dirichlet boundary conditions.

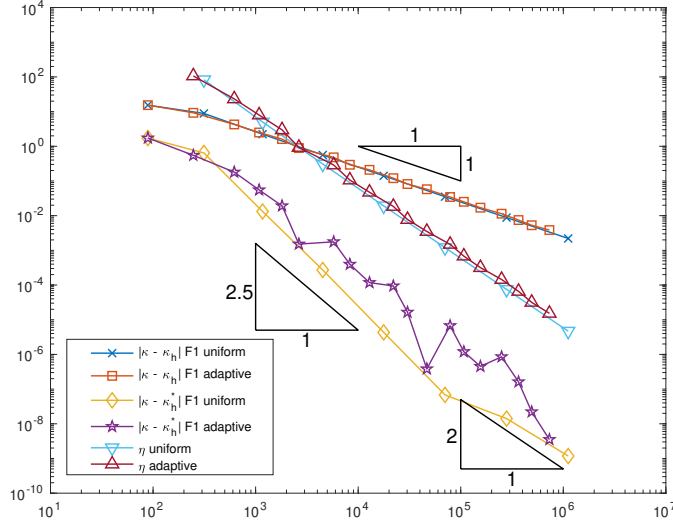


Figure 8: Convergence history for the third eigenvalue and the estimator using the lowest-order Falk element, $k = 1$, on the square with Dirichlet boundary conditions.

We consider homogeneous Dirichlet boundary conditions, i.e.

$$u = 0 \quad \text{on } \partial\Omega.$$

In the following discussion, let us assume that κ_h^* is positive. By Lemma 5.6, the following identity is valid:

$$\begin{aligned} \kappa - \kappa_h^* = & \|\mathbb{C}^{-1}(\sigma - \sigma_h) + (\gamma - \gamma_h)\|_{\mathbb{C}^{1/2}}^2 - 2\mu\|\gamma - \gamma_h\|_0^2 + \kappa_h^*\|u - u_h^*\|_0^2 \\ & + 2(\operatorname{div}(\sigma - \sigma_h), u - u_h^*) - 2((\kappa_h^* - \kappa)u, u - u_h^*). \end{aligned} \quad (69)$$

In the lowest-order Falk finite element approximation, $k = 1$, the postprocessed Falk finite element eigenfunction approximation is one order less, and of the same order as the stress approximation and the skew-symmetric part approximation. By Remark 5.8, we have for $k = 1$ sub-optimal postprocessed eigenvalue approximation

$$|\kappa - \kappa_h^*|_0 = \mathcal{O}(h^3).$$

We would therefore also expect this in the numerical results. However, this is not the case and we observe in all of our experiments a better convergence rate in the lowest-order case $k = 1$. In fact, in Figure 8 we observe pre-asymptotically an even higher convergence rate than four.

There are indeed several phenomena that happen here. First, the analysis does not provide a statement whether the $2(\operatorname{div}(\sigma - \sigma_h), u - u_h^*)$ is positive or negative. If it were negative, we could neglect it and would therefore expect fourth-order convergence in the postprocessed eigenvalue approximation. If, on the other hand, it were not negative, we would achieve only third-order convergence. According to the numerical results, it must therefore be negative. However, this cannot be deduced from the analysis and we must assume that the estimate in Remark 5.8 is not rigorous.

Second, the superconvergence we observe in Figure 8 can be explained as follows. The last term in (69) is of negligible for h sufficiently small. The penultimate term must also negligible due to the observations before. The first three terms in

$$\|\mathbb{C}^{-1}(\sigma - \sigma_h) + (\gamma - \gamma_h)\|_{\mathbb{C}^{1/2}}^2 - 2\mu\|\gamma - \gamma_h\|_0^2 + \kappa_h^*\|u - u_h^*\|_0^2 \quad (70)$$

all have convergence order four. Superconvergence is only possible if the fourth-order terms in (70) cancel each other, for example due to symmetry in the mesh. Therefore, the superconvergence phenomena might vanish on adaptively refined meshes, because we cannot expect cancellation to happen.

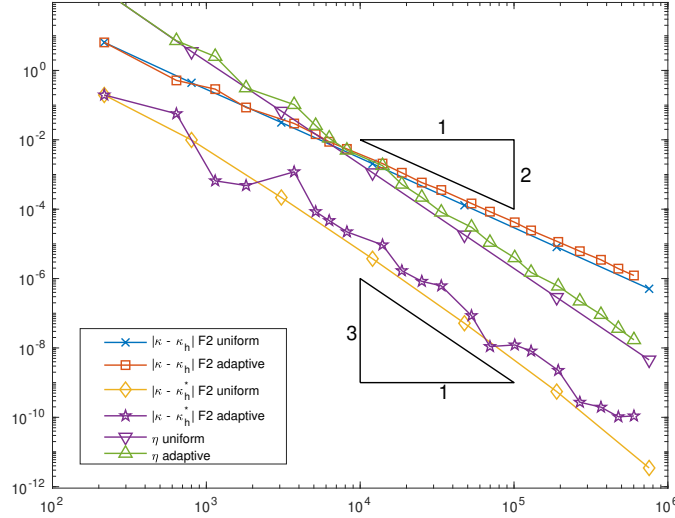


Figure 9: Convergence history for the third eigenvalue and the estimator using the second-order Falk element, $k = 2$, on the square with Dirichlet boundary conditions.

Figure 9 shows that there is no superconvergence for the second-order Falk finite element postprocessed eigenvalue approximation, $k = 2$. On both, uniformly and adaptively refined meshes, we observe optimal order of convergence. We know that $\|u - u_h^*\|_0^2$ is of higher order than $\|\sigma - \sigma_h\|_0^2$ and $\|\gamma - \gamma_h\|_0^2$ and cancellation does not happen.

The postprocessed third-order Falk finite element eigenvalue approximation shows reduced order of convergence on the uniformly refined mesh, c.f. Figure 10. On an adaptively refined mesh, the optimal order of convergence is recovered. In the post-processing, we assumed that the eigenfunction u is regular. The absence of optimal order of convergence on the uniformly refined mesh can be explained by the regularity limit of the solution. The error estimator η exceeds the eigenvalue approximation error for both uniformly refined meshes and adaptively refined meshes and approaches the postprocessed eigenvalue approximation error from above.

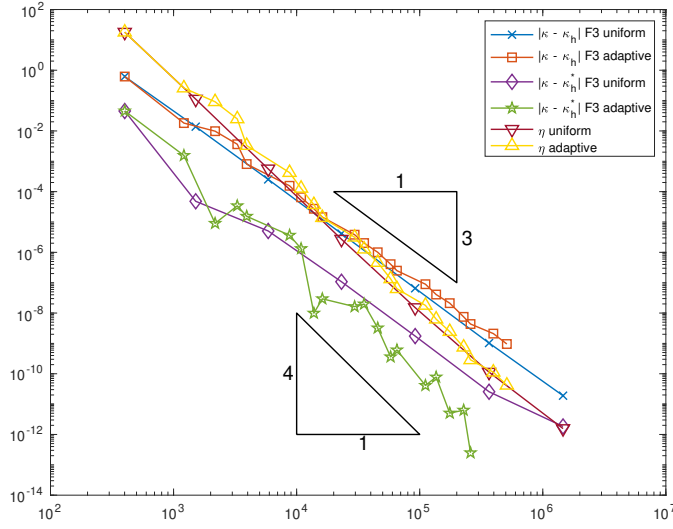


Figure 10: Convergence history for the third eigenvalue and the estimator using the third-order Falk element, $k = 3$, on the square with Dirichlet boundary conditions.

Neumann boundary conditions

The pure Neumann boundary problem corresponds to

$$\sigma \cdot n = g \quad \text{on } \partial\Omega$$

and let $g = 0$ in the following. The eigenvalues in the spectrum for which we know the exact value for different Lamé parameters λ and μ on the square domain with imposed Neumann boundary conditions are displayed in Table 1. Interestingly, the known eigenvalues appear at different positions in the spectrum depending on the parameter λ . For example, the seventh eigenvalue for $\lambda = 1$ and the fifth for $\lambda = 10$, $\lambda = 100$ coincide. Due to rigid body motions, the first three eigenvalues in the spectrum equal to zero.

$\lambda = 1$	$\lambda = 10$	$\lambda = 100$	
Eigenvalue nr.	Eigenvalue nr.	Eigenvalue nr.	Value
1	2	3	0
1	2	3	0
1	2	3	0
7	5	5	$2\pi^2\mu$
16	13	13	$8\pi^2\mu$
29	25	25	$18\pi^2\mu$
49	42	39	$32\pi^2\mu$
69	57	56	$50\pi^2\mu$
95	81	75	$72\pi^2\mu$
123	103	97	$98\pi^2\mu$
161	131	125	$128\pi^2\mu$
197	163	155	$162\pi^2\mu$

Table 1: Some known eigenvalues on the square domain with homogeneous Neumann boundary conditions.

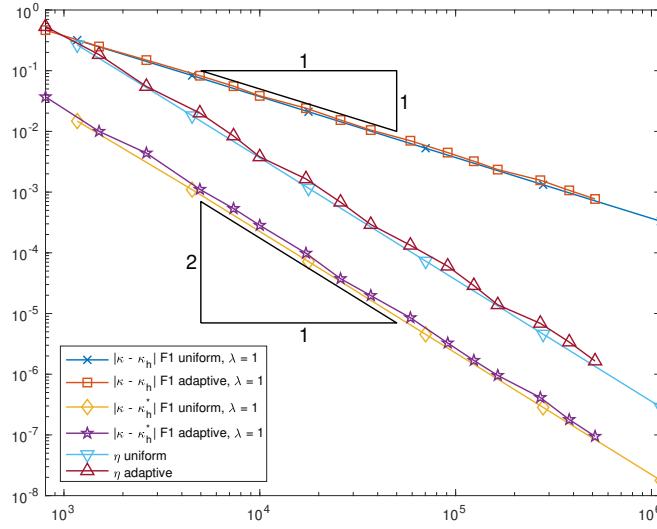


Figure 11: Convergence history of the seventh eigenvalue approximation error and the estimator for the lowest-order Falk element on the square with homogeneous Neumann boundary conditions.

Similar to the square domain with homogeneous Dirichlet boundary conditions, the square domain with homogeneous Neumann boundary conditions, cf. Figure 11, reveals a better convergence rate than we would expect for the lowest-order Falk finite element approximation on both uniformly and adaptively refined meshes. The discussed superconvergence phenomena does not appear here and the error estimator η shows good results.

As the eigenvalue number in the spectrum increases, the postprocessed eigenvalue approximation error and the error estimator become larger at the same order of convergence, c.f. Figure 12 for the seventh and the sixteenth eigenvalues in the spectrum.

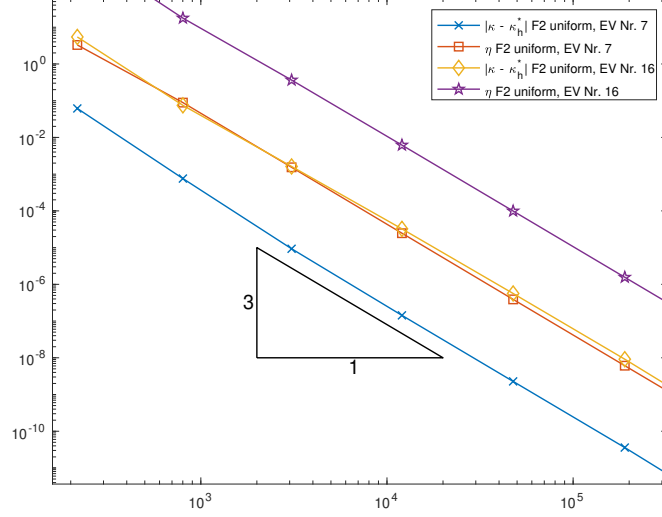


Figure 12: Convergence history of the seventh and the sixteenth postprocessed eigenvalue approximation error and the estimator for the second-order Falk element on the square with homogeneous Neumann boundary conditions.

Symmetric boundary conditions

Let n be the outward pointing unit normal and t be a unit tangential. Then symmetric boundary conditions are

$$\begin{aligned} t^\top \cdot \sigma \cdot n &= 0 & \text{on } \partial\Omega, \\ u \cdot n &= 0 & \text{on } \partial\Omega. \end{aligned}$$

For symmetric boundary conditions, by Domínguez, Nilima and Sun [15], we know the exact values of the eigenpairs

$$(\kappa_s^2, u_s) \in \mathbb{R} \times U \quad (\kappa_p^2, u_p) \in \mathbb{R} \times U.$$

On a rectangle $[0, a] \times [0, b]$, these are given by

$$\begin{aligned} u_s &:= \begin{pmatrix} (al) \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{l\pi y}{b}\right) \\ -(bm) \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{l\pi y}{b}\right) \end{pmatrix}, \\ \kappa_s^2 &:= \mu\pi^2(m^2 + l^2), \end{aligned}$$

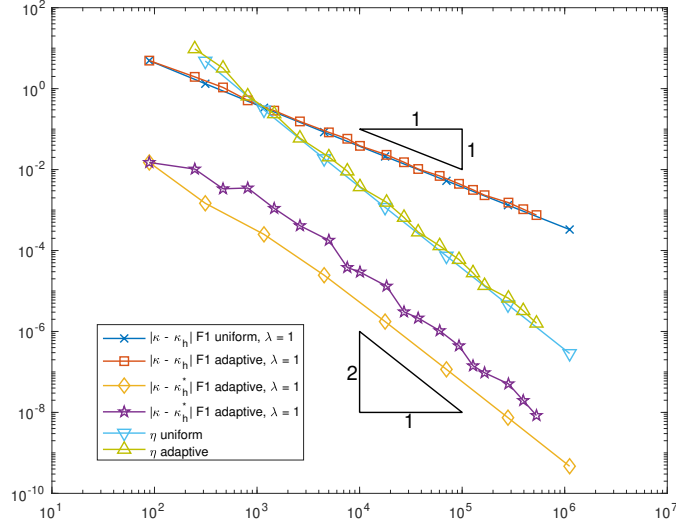


Figure 13: Convergence history of the postprocessed lowest-order Falk finite element eigenvalue approximation and the estimator on the square domain with symmetric boundary conditions.

for $m, l \in \mathbb{N}$ and

$$u_p := \begin{pmatrix} (bm) \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{l\pi y}{b}\right) \\ (al) \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{l\pi y}{b}\right) \end{pmatrix},$$

$$\kappa_p^2 := (\lambda + 2\mu)\pi^2(m^2 + l^2),$$

for $m, l \in \mathbb{N}_0$ with $m + l > 0$. They are called s -modes and p -modes respectively. In our numerical experiments, we are considering symmetric boundary conditions on the unit square, i.e. $a = b = 1$.

One may observe $\operatorname{div} u_s = 0$ and $\operatorname{curl} u_p = 0$.

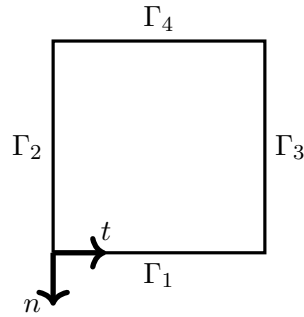


Figure 14: Unit Square.

Symmetric boundary conditions on the square

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}^\top \cdot \sigma \cdot \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}^\top \cdot \sigma \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\sigma_{12} \quad \text{on } \Gamma_1 \cup \Gamma_4,$$

$$u \cdot \begin{pmatrix} 0 \\ -1 \end{pmatrix} = -u \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -u_2 \quad \text{on } \Gamma_1 \cup \Gamma_4,$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix}^\top \cdot \sigma \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}^\top \cdot \sigma \cdot \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \sigma_{21} \quad \text{on } \Gamma_2 \cup \Gamma_3,$$

$$u \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = -u \cdot \begin{pmatrix} -1 \\ 0 \end{pmatrix} = u_1 \quad \text{on } \Gamma_2 \cup \Gamma_3,$$

fix one component of each u and σ on the boundary segments $\Gamma_1 \cup \Gamma_4$ and $\Gamma_2 \cup \Gamma_3$. The situation is different for pure Dirichlet boundary conditions on the square domain, where both components of u are fixed, and for pure Neumann boundary conditions on the square domain, where the two non-diagonal components of σ are fixed on every boundary segment.

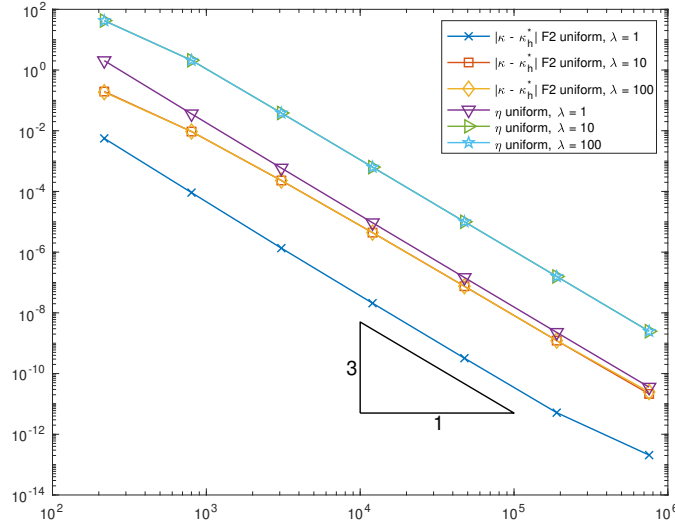


Figure 15: Convergence history of the postprocessed second-order Falk finite element approximation of the eigenvalue and the estimator for different parameter $\lambda = 1$, $\lambda = 10$ and $\lambda = 100$.

It is well-known that the solution can degenerate when λ tends to infinity. This phenomenon is called Locking. It may be noted that the classical pure displacement approach in elasticity leads to Locking in the approximation of the displacement for various finite elements. Furthermore, the stress is subsequently calculated using

$$\sigma = \mathbb{C}\epsilon(u)$$

and it is clear that this also leads to Locking in the approximation of the stress. For

the pure displacement eigenvalue problem, Locking would also occur in the eigenvalue approximation.

In our formulation, only $\mathbb{C}^{-1}$ appears and by Remark 5.4, we do not expect Locking to happen. For increasing λ , the postprocessed eigenvalue approximation error and its estimator η do not degenerate. The errors may slightly vary for small variations in λ but asymptotically (in λ) they do not. This can numerically be observed in Figure 15.

7.2 L-shaped domain

For the L-shaped domain $\Omega = (-1, 1) \times (0, 1) \cup (-1, 0) \times (-1, 0)$ with homogeneous Dirichlet boundary conditions on the entire boundary $\partial\Omega = \Gamma_D$, the exact eigenvalues are unknown. As a reference value, we choose $\kappa = 13.591920213185$ for the smallest eigenvalue in the spectrum. The L-shaped domain is non-convex and has a re-entrant corner at the origin.

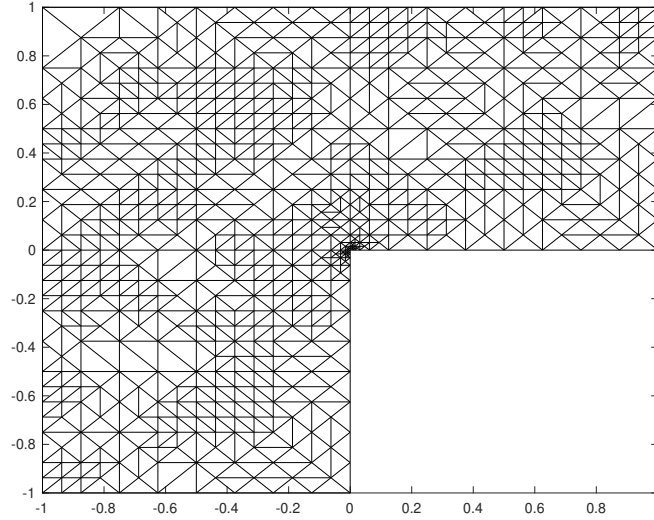


Figure 16: Adaptive mesh refinement for $\min NrDoF = 1e4$ of the the L-shaped domain using the lowest-order Falk element.

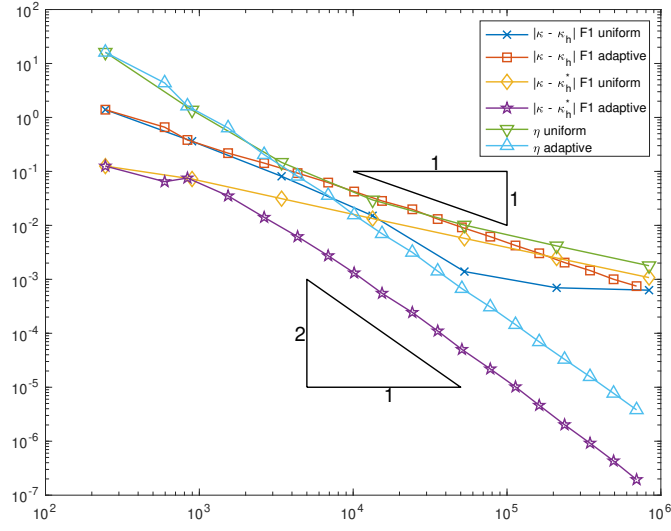


Figure 17: Convergence history of the eigenvalue approximation error, the postprocessed eigenvalue approximation error and the estimator η for the lowest-order Falk element on the L-shaped domain.

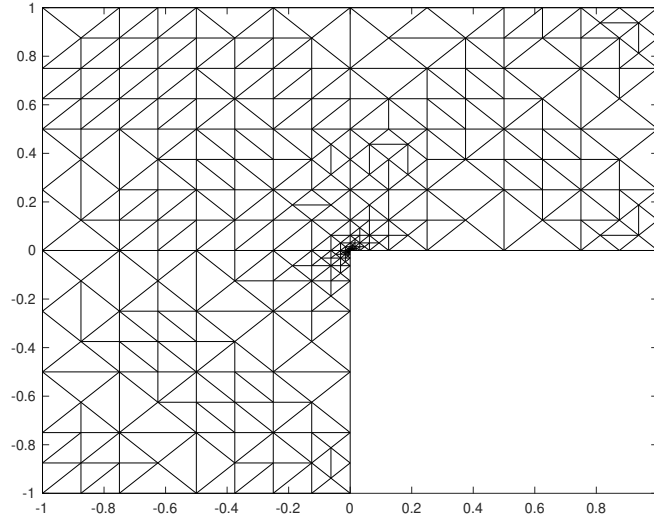


Figure 18: Adaptive mesh refinement for $\min NrDoF = 1e4$ of the the L-shaped domain using the second-order Falk element.

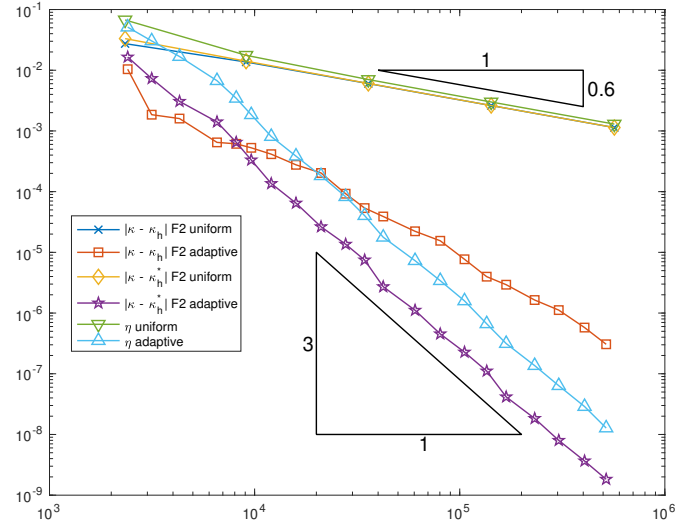


Figure 19: Convergence history of the eigenvalue approximation error, the postprocessed eigenvalue approximation error and the estimator for the second-order Falk element on the L-shaped domain.

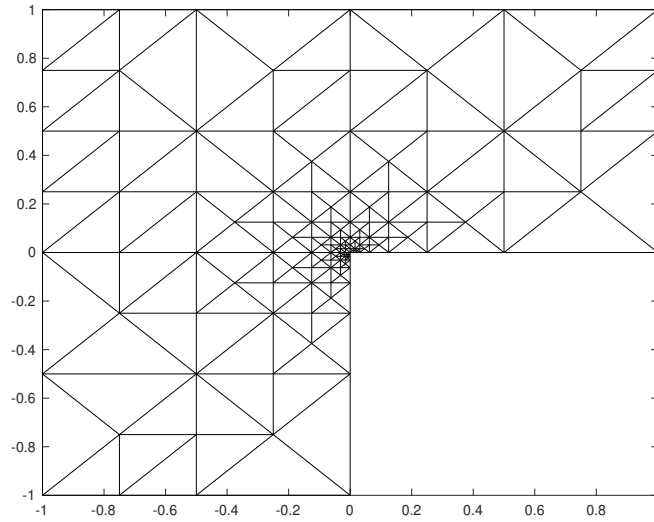


Figure 20: Adaptive mesh refinement for $\min NrDoF = 1e4$ of the the L-shaped domain using the third-order Falk element.

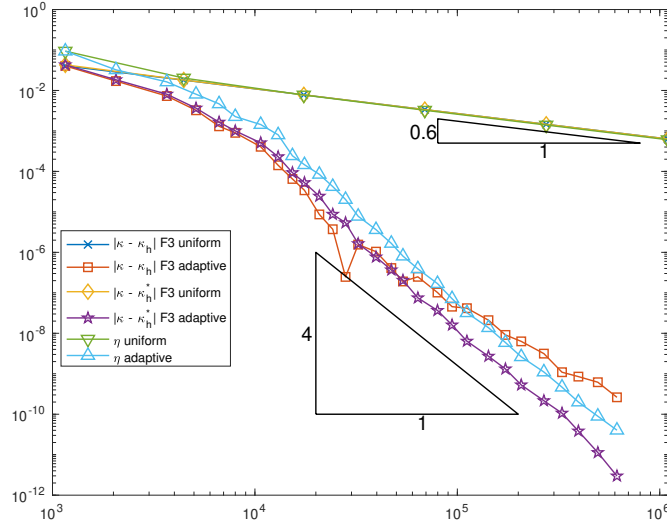


Figure 21: Convergence history of the eigenvalue approximation error, the postprocessed eigenvalue approximation error and the estimator for the third-order Falk element on the L-shaped domain.

On the L-shaped domain, we cannot expect solutions of (36) to be regular. But in the postprocessed Falk finite element eigenfunction approximation error analysis, we have assumed regularity. Therefore, the results in Figure 17, Figure 19, and Figure 21 show a reduced order of convergence for the eigenvalue approximation and the error estimator η on uniformly refined meshes. Adaptive mesh refinement almost recovers optimal order of convergence even for third-order Falk finite element approximations.

Figure 16, Figure 18, and Figure 20 show adaptively refined meshes using the error estimator η . The figures reveal that more refinement is happening at the re-entrant corner, which increases for higher-order Falk finite element approximations.

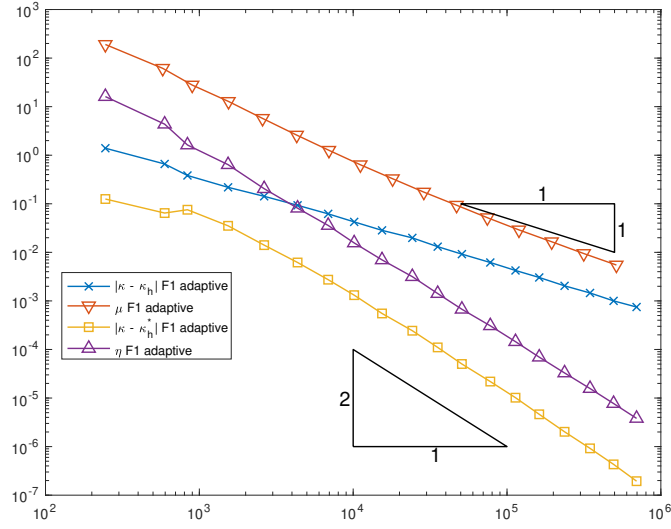


Figure 22: Convergence history of the eigenvalue approximation error and the estimators η and μ for the lowest-order Falk element on the L-shaped domain.

In Figure 22, the error estimator μ for the eigenvalue approximation error and the error estimator η for the postprocessed eigenvalue approximation error are displayed for the lowest-order Falk element on adaptively refined meshes. Both error estimators show good results.

7.3 Cook's membrane

Cook's membrane benchmark is a tapered panel, that is clamped on the left side at the boundary and subjected to an upward shear force in vertical direction on the right side at the boundary. This problem is a standard benchmark test for bending dominated response.

The corresponding boundary conditions are

$$\begin{aligned} u &= 0 & \text{on } \Gamma_D &= \{(x, y) \in \bar{\Omega} : x = 0\}, \\ \sigma \cdot n &= g & \text{on } \Gamma_N &= \{(x, y) \in \bar{\Omega} : x = 48\}. \end{aligned}$$

In this problem, there are no applied volume forces acting on the elastic body, i.e. $f(x, y) = 0$ and the surface traction is $g(x, y) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ on Γ_N and $g(x, y) = 0$ on $\partial\Omega/\Gamma_N$.

As a reference eigenvalue, we choose $\kappa = 0.0004029679680725$. Figure 24 shows again the better convergence rates in the lowest-order Falk finite element approximation. On uniformly refined meshes, the eigenvalue approximation errors and the error estimator reveal a reduced order of convergence, which again can be explained by the regularity limit of the eigenfunction u .

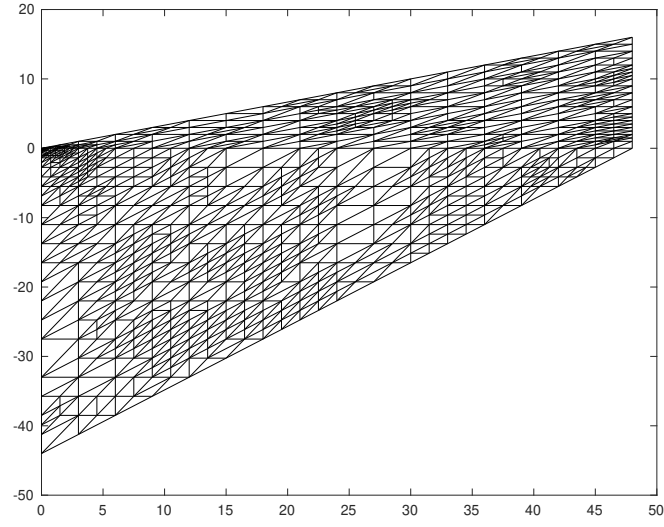


Figure 23: An adaptively refined mesh of Cook's membrane benchmark using the error estimator η for the lowest-order Falk element.

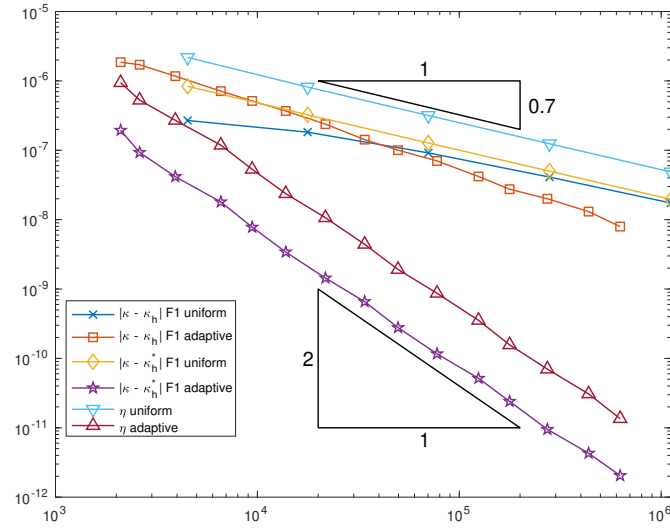


Figure 24: Convergence history of the lowest-order Falk finite element eigenvalue approximation and the error estimator.

7.4 Conclusions and outlook

The presented postprocessing for the stable Falk finite element approximation of the eigenvalue problem in elasticity with weakly imposed symmetry is numerically shown to be very useful. Additionally, this method does not cause Locking. The numerical experiments in the lowest-order case present even better results than we would expect from the analysis. Moreover, in the lowest-order case we observe the extraordinary phenomenon of superconvergence due to cancellation in some examples and on some meshes. The error estimator η for the postprocessed eigenvalue approximation is numerically shown to be reliable and efficient. On domains with singularities, the use of η for an adaptive refinement of the mesh almost recovers optimal order of convergence. For the pure Neumann problem on the square domain, some exact eigenvalues could be found.

In a further discussion, it would be interesting to study other stable elements. For example, elements that are additionally stabilised using bubble functions inside the elements. An overview of known elements is given by [21]. Here, it would be interesting to investigate whether the phenomenon of superconvergence also appears or if it occurs exclusively in the approximation using the Falk element. Besides, it may be interesting to analyse how the condition of the matrix influences the eigenvalue approximation for unknown eigenvalues and on domains with mixed boundary conditions.

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